Service Design

SafeCyl - LPG Monitoring Service

Assessment of Advanced Proficiency

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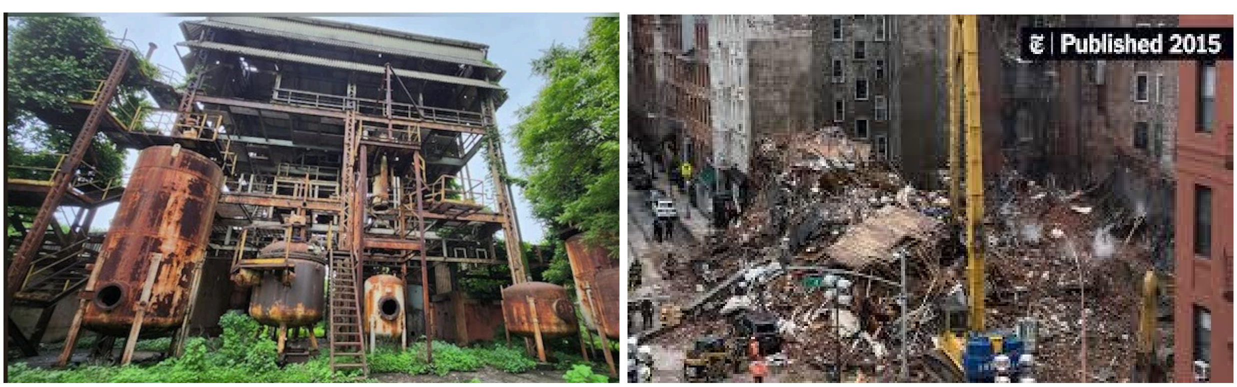
Introduction

Background and Context

The management of liquefied petroleum gas (LPG) cylinders represents a fundamental challenge in Indian urban households, where traditional practices intersect with modern safety requirements. Despite India's remarkable achievement in providing LPG access to over 290 million households through initiatives like the Pradhan Mantri Ujjwala Yojana, the service ecosystem surrounding LPG management remains largely unchanged since its introduction decades ago. This stagnation in service innovation occurs against a backdrop of rapid technological advancement in other household utilities, creating a stark contrast between how Indians manage their electricity, water, and gas resources.

The current LPG management paradigm operates on reactive principles, where users engage with the service only at moments of crisis; when gas exhausts unexpectedly, when a leak is suspected, or when delivery delays disrupt meal preparation. This reactive approach generates continuous anxiety among users, who resort to primitive methods like physically lifting cylinders or tapping them to estimate remaining gas. The absence of visibility into consumption patterns forces households to maintain expensive backup cylinders or face the inconvenience of emergency restaurant orders when gas depletes during cooking. Safety concerns amplify these operational challenges. The Petroleum and Natural Gas Regulatory Board reported 3,068 LPG-related incidents in 2023, with 42% attributed to undetected leaks. The human olfactory system, which serves as the primary leak detection mechanism in most households, requires 3-5 minutes to detect gas concentrations that electronic sensors identify within 30 seconds. This detection delay creates a critical window of vulnerability where gas accumulation can reach dangerous levels, particularly in poorly ventilated urban kitchens where space constraints often compromise safety protocols.



Infrastructure Damage Caused By Gas Leaks and Explosions (India, 2015 - 2024)

Problem Statement

Problem Significance

The implications of inadequate LPG management extend beyond individual households to encompass broader societal costs. Economic losses from inefficient LPG management manifest in multiple ways:

- Emergency cylinder booking premiums (₹50-100 per incident)

- Unexpected restaurant expenses when gas exhausts during cooking (₹300-500 per occurrence)

- The opportunity cost of maintaining backup cylinders (₹1,500 security deposit locked unnecessarily).

When aggregated across urban households experiencing these issues, the cumulative economic impact reaches significant proportions. Households face additional expenses of ₹500-800 annually due to poor visibility and planning capabilities, which extrapolates to approximately ₹5,000 crores nationally when considering 100 million urban households. Beyond direct economic costs, undetected micro-leaks represent both safety hazards and resource wastage. While major leaks trigger sooner detection, minor leaks below the odor threshold can persist unnoticed, releasing gas that serves no useful purpose while depleting cylinder contents faster than expected. Additionally, improper regulator closure after cooking sessions, particularly common among elderly users with reduced grip strength, contributes to gradual gas loss that accumulates over time. Moreover, the service delivery infrastructure exhibits systematic failures in safety compliance. Legal mandates require delivery personnel to check cylinder weight and test for leaks in the customer's presence, yet our research indicates these checks occur in only 10% of deliveries. This non-compliance removes a crucial safety checkpoint from the service chain, transferring risk entirely to end users who lack the knowledge, tools, or confidence to perform adequate safety assessments.

Initial Problem Statement

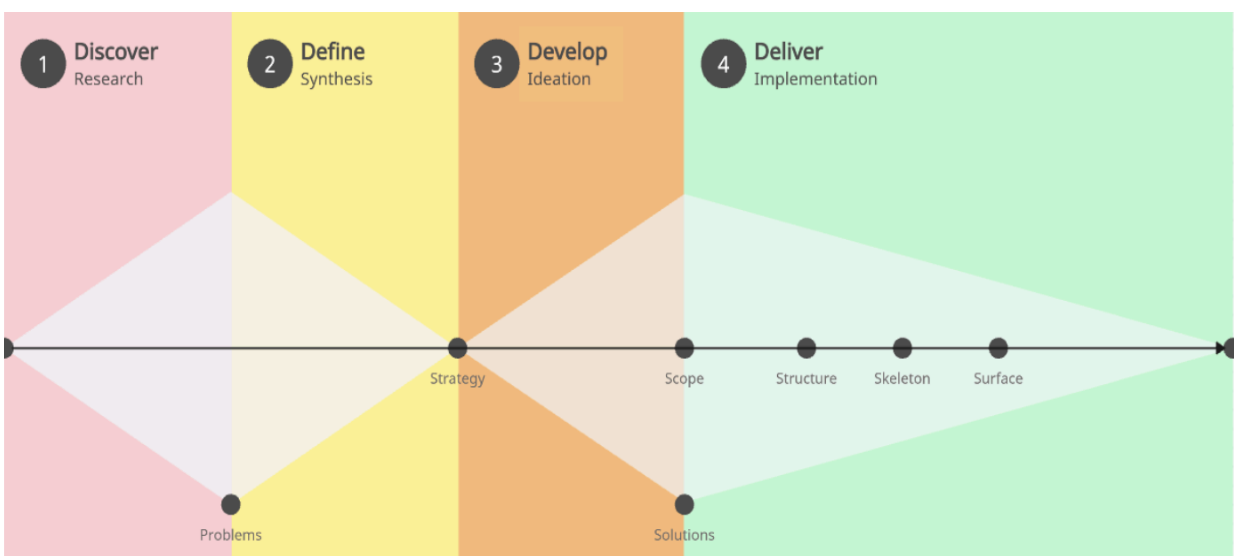
Based on preliminary observations of the LPG management landscape in urban India and initial rounds of contextual inquiry, our initial problem statement is:

"How might we transform LPG cylinder management from a reactive, anxiety-driven crisis management system into a predictable, safe utility service that provides users with the operational visibility, safety verification, and planning capability they currently lack?"

This problem statement will now be tested, refined, and crystallized through comprehensive primary research; including quantitative surveys to measure the scope and frequency of pain points, qualitative interviews to understand emotional dimensions and behavioral patterns, and stakeholder analysis to map ecosystem dynamics. The following research phase will either validate this initial hypothesis or reveal deeper systemic issues requiring reframing, ensuring our final problem statement is grounded in empirical evidence rather than assumptions.

Methodology Overview

The project employed an adapted double diamond design framework integrated with Porter & Heppelmann's smart connected product development methodology, progressing through four phases: Discover, Define, Develop, and Deliver. This structured approach ensured comprehensive problem exploration before technical solution development while maintaining focus on the evolution of product capabilities from monitoring to autonomy.



Phase 1: Discover (Research & Understanding

Objective

Develop comprehensive, evidence-based understanding of current LPG management service ecosystem, user behaviors, pain points, and systemic failures through primary research and stakeholder analysis.

User Research Component

Quantitative Research (n=50):

Method: Structured digital survey with 20 questions across 5 thematic domains

- Demographic profiling
- Current LPG management practices and behaviors
- Safety awareness levels and detection capabilities
- Service experience and compliance observations
- Technology adoption readiness and price sensitivity

Sampling: Urban households across Bangalore (35%), Pune (40%), Mumbai (15%), and other metros (10%). Distribution: Resident welfare associations, professional networks, social media platforms

Qualitative Research (n=3):

Method: Semi-structured in-depth interviews (45-60 minutes each)

Protocol: Four thematic sections exploring LPG routines, critical incidents, safety perceptions, and ideal future states

Participant Selection: Purposive sampling representing three distinct user archetypes:

- Safety-anxious working parent (representative of nuclear families, high safety concern)

- Convenience-seeking young professional (representative of singles, reactive behavior)

- Accessibility-challenged elderly user (representative of physical limitations, low digital literacy)

Phase 2: Define (Synthesis & Architecture Design)

Problem Synthesis

- Research synthesis through storyboarding, persona development, and journey mapping

- Problem statement crystallization based on identified pain points and technical opportunities

Root Cause Analysis:

- Applying 5 Whys methodology to major service failures
- Distinguishing symptoms from root causes
- Identifying systemic issues vs. individual user behaviors

Current Service Ecosystem Map:

- Visual representation of actors, resources, touchpoints, and flows
- Identification of information asymmetries and broken connections
- Mapping regulatory environment and compliance gaps
- Highlighting perverse incentive structures and misalignments

Persona Development (3 Archetypes)

- Priya (35) - The Safety Guardian: Working mother, high safety anxiety, maintains backup cylinder

- Arjun (28) - The Convenience Seeker: Single professional, reactive behavior, tech-enthusiast

- Ramesh (65) - The Accessibility Challenger: Retired, physical limitations, low tech literacy

Customer Journey Mapping (Current State - As-Is):

- Pre-use → Active use → Near exhaustion → Crisis → Booking → Delivery
- Customer actions and behaviors
- Thoughts and internal dialogue
- Emotional states (visualized as emotion curve
- Pain points and frustration triggers

- **Touchpoint Identification**: Physical (cylinder, regulator), Digital (booking app), Human (delivery agent) and **Opportunity Spotting**: Moments where service intervention could prevent crisis

Future-State Service Blueprint

- Redesigned touchpoints incorporating SafeCyl device, mobile app, digital verification

- Transformed backstage processes: predictive analytics, automated booking, compliance dashboards

- New support infrastructure: Firebase cloud, ESP32 edge processing, SaaS distributor tools

Value Proposition Canvas:

- Customer Profile: Jobs to be done (functional, social, emotional), Pains, Gains
- Value Map: Products & Services, Pain Relievers, Gain Creators
- Fit Assessment: How SafeCyl addresses each customer need across three personas

Phase 3: Develop (Prototype & Iteration)

Hardware Development

- Iterative prototyping from breadboard to integrated PCB
- Sensor calibration and accuracy optimization
- Enclosure design for kitchen environment durability

Software Development

- Firmware development for ESP32 using Arduino IDE
- Firebase Realtime Database implementation
- Mobile application development using MIT App Inventor
- Predictive algorithms for consumption forecasting
- Integration testing across all system components

Phase 4: Deliver (Validation & Business Model)

Technical Validation

- Performance metrics: response time, accuracy, reliability
- Safety compliance verification against LPG regulations

Business Development

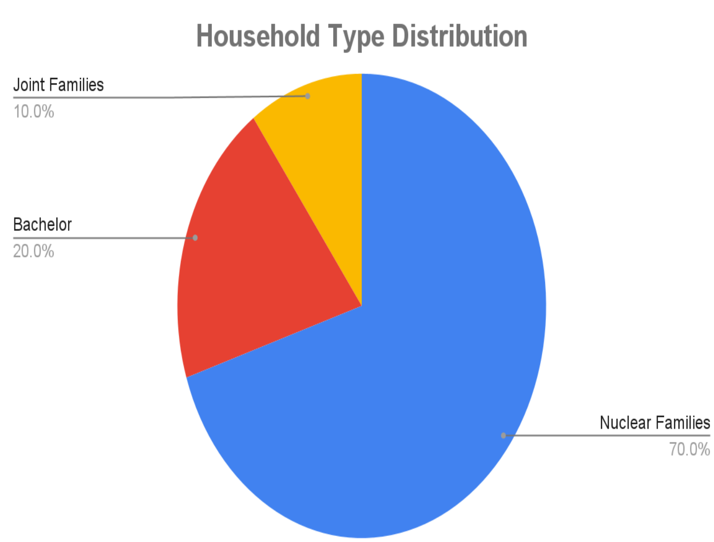
- Unit economics refinement from prototype (₹1,920) to production (₹1,200-1,300)
- Three-phase go-to-market strategy development
- Partnership framework with gas agencies
- System-of-systems roadmap for ecosystem evolution

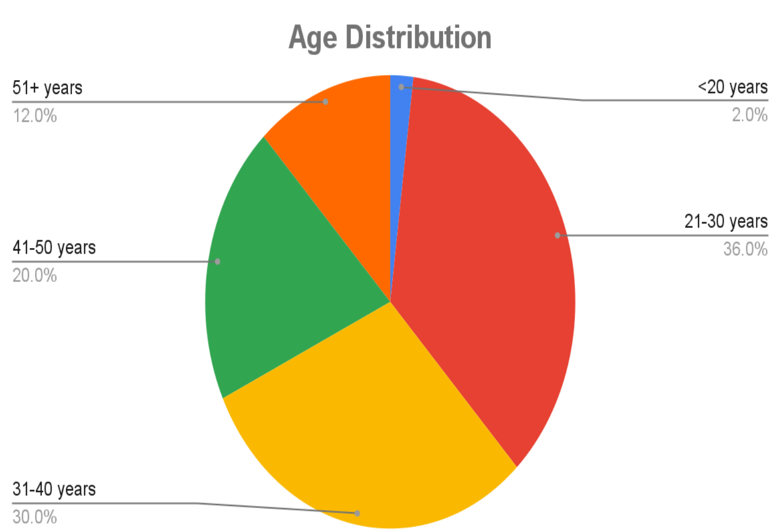
This methodology ensured the project progressed systematically from understanding user needs through technical implementation to a validated service design business model, with continuous iteration

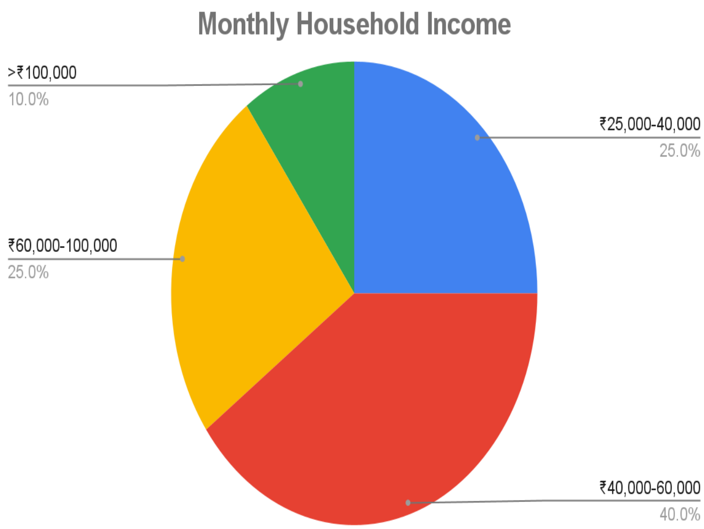
Research Methodology and Execution

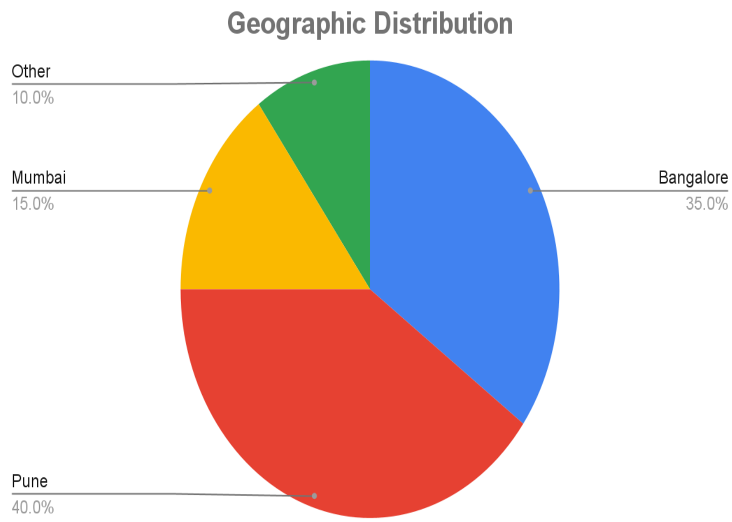
Sample Demographics

The research sample achieved representation across key demographic variables relevant to LPG management:







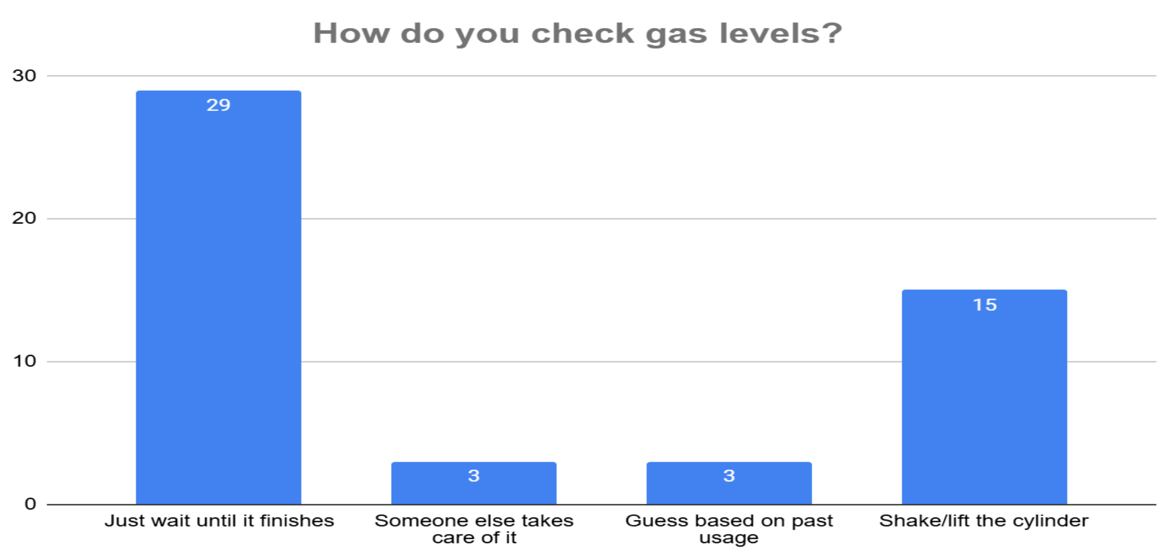


Research Findings and Analysis

<u>Current State Behaviors</u>

Analysis of quantitative data revealed concerning patterns in current LPG management practices:

Gas Level Monitoring Methods



Key Findings

1. **Dominant Reactive Approach (58%)**
   The majority of respondents (29 out of 50) employ a completely passive strategy, waiting until gas exhaustion before taking action. This represents a critical service failure where nearly 6 in 10 users operate without any proactive monitoring system.

2. **Physical Checking Methods (30%)**
   The second-largest group relies on physically lifting or shaking cylinders to estimate remaining gas. This method is:

   - Physically demanding (especially problematic for 12% of respondents who are 51+ years)

   - Highly inaccurate (±2-3 kg margin of error)

   - Not feasible for all users

3. **Dependency and Guesswork (12% combined)**
   Six respondents either delegate responsibility entirely (6%) or attempt estimation based on historical patterns (6%). Both approaches lack reliability and create vulnerability to unexpected exhaustion.

Critical Insights

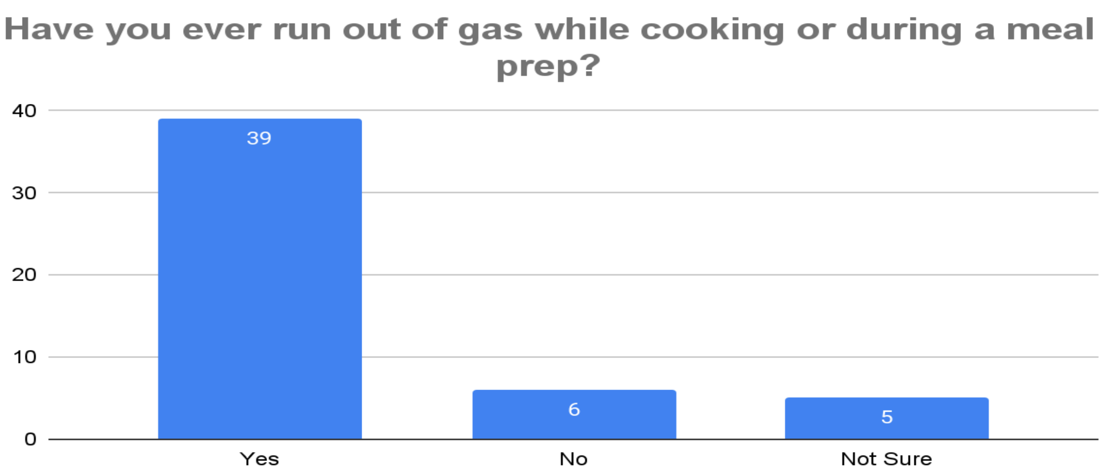
- **88% lack reliable monitoring** -
  Combining those who wait (58%), shake cylinders (30%), this vast majority has no accurate, consistent method for gas level awareness.

- **Complete information darkness** -
  The 58% who "just wait" operate with zero visibility into their gas consumption, making planning impossible and anxiety inevitable.

This data strongly validates the problem statement, showing that the current LPG management system fails to provide basic visibility that users need for informed decision-making.

"When gas finishes while cooking dinner, it's not just about ordering food. It's the stress, the waste of half-cooked food, the family disappointment, and then the anxiety about when it will happen again." - ***Homemaker, Age 35***

The Exhaustion Crisis



<u>Key Findings</u>

1. **Near-Universal Service Failure (78%)**
   An overwhelming 39 out of 50 respondents have experienced gas exhaustion during cooking. This isn't an occasional inconvenience—it's a predictable system failure affecting nearly 4 out of 5 households. The impact cascades beyond the immediate meal:

   - Wasted ingredients from half-cooked food (average ₹200-300 per incident)
   - Emergency restaurant orders (₹400-600 for family of four)
   - Time lost in crisis management (45-60 minutes)
   - Emotional stress on family dynamics

2. **The "Lucky" Minority (12%)**
   Only 6 respondents have never experienced exhaustion. Further investigation reveals these users either:

   - Maintain expensive backup cylinders (₹1,500 deposit locked)
   - Have household help managing multiple cylinders

- Use minimal gas due to frequent eating out

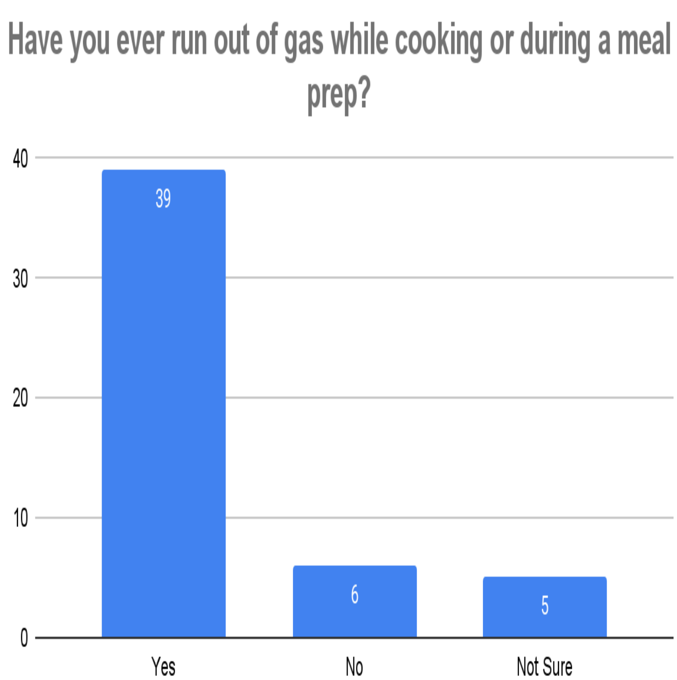
3. **Uncertainty as Normalization (10%)**
   5 respondents "aren't sure" if they've run out—suggesting the experience is so routine they cannot distinguish specific incidents from general pattern.

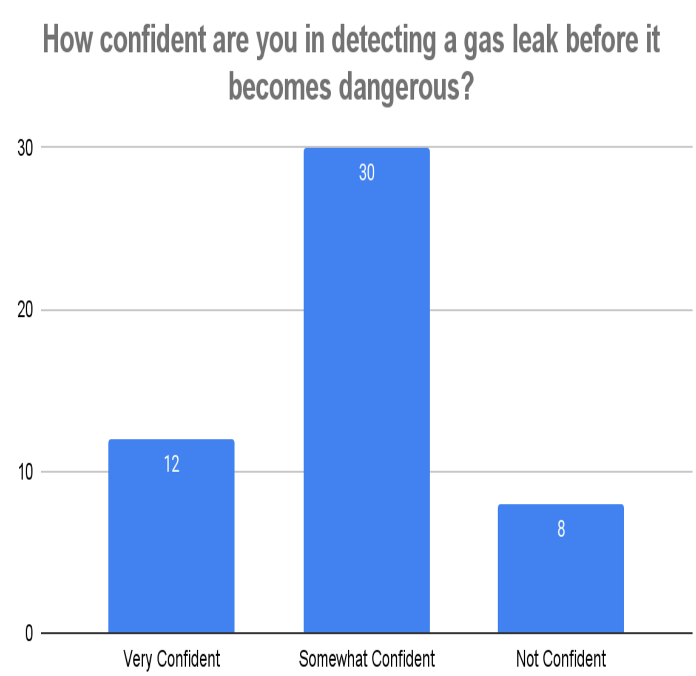
Critical Insight

The 78% exhaustion rate, when mapped against the 58% who "just wait until it finishes," reveals a vicious cycle: reactive management guarantees crisis, yet users lack tools for proactive approach.

"Last month it happened during my daughter's birthday dinner. Twelve guests, half-cooked biryani, and me calling every restaurant that could deliver for 15 people. We don't talk about that evening." - ***Software Engineer, Age 34***

The Safety Blindspot





Key Findings

1. **Widespread Leak Suspicion (72%)**
   36 out of 50 respondents have suspected gas leaks. This high incidence suggests either:

   - Actual micro-leaks are common (indicating equipment/maintenance issues)
   - Users cannot distinguish between normal gas smell during connection changes versus actual leaks
   - Anxiety manifests as hypervigilance to any gas odor

2. **Dangerous Confidence Gap**
   Breaking down detection confidence against actual capability

   - "Very confident" (24%): 12 respondents believe they can detect leaks early
   - Reality check: Only 7 respondents (14%) know the soapy water test
   - The gap: At least 5 "very confident" users have no actual detection method

   This overconfidence is lethal. Users believing they'll smell gas don't realize:

   - LPG needs 1.5-2% concentration for odor detection
   - Explosive range starts at 2.1% concentration
   - The safety margin is essentially zero

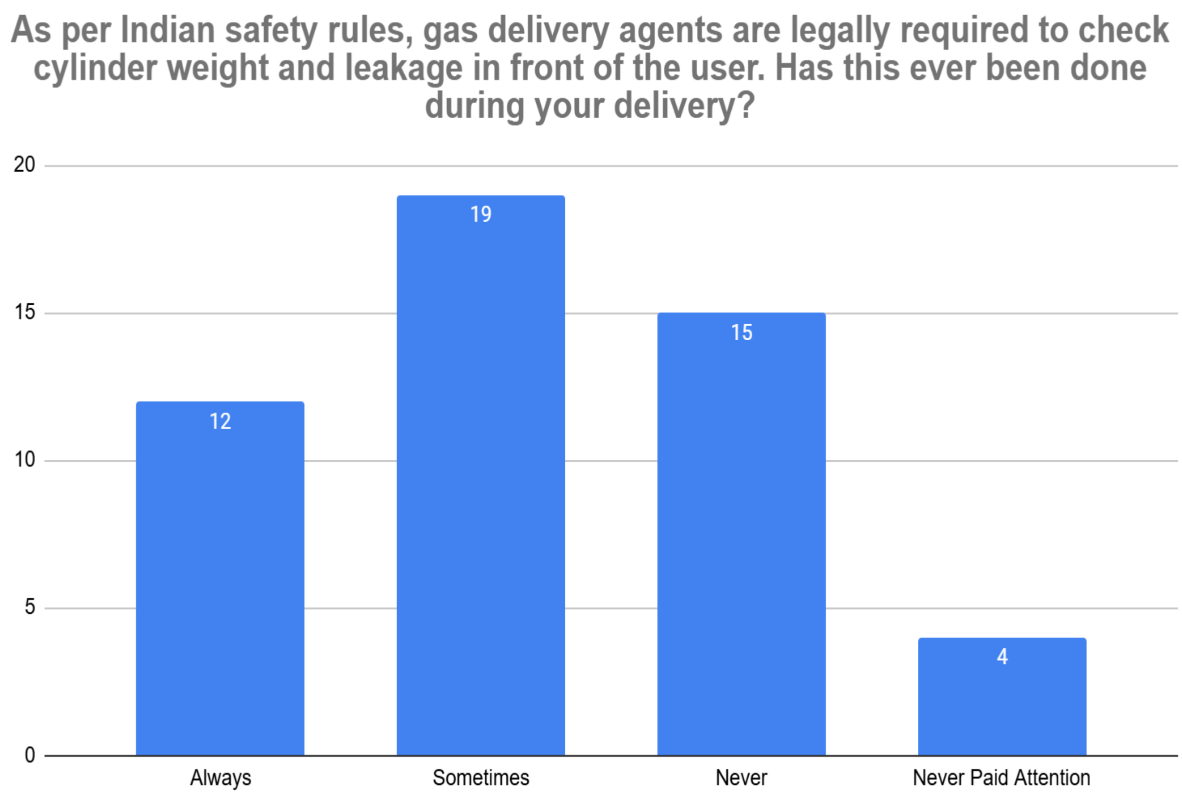
3. **The Honest Middle (60%)**
   30 respondents admit being only "somewhat confident", a more realistic assessment given their reliance on smell alone. These users experience constant low-grade anxiety, sniffing for gas but never certain of their assessment accuracy.

4. **The Acknowledged Vulnerable (16%)**
   8 respondents admit they're "not confident" in detection, honest self-assessment that paradoxically makes them safer through acknowledged caution.

Critical Insight

The intersection of 72% leak suspicion with only 14% knowing proper detection creates a population living with unverified fear, they suspect danger but cannot confirm safety.

Service Delivery Breakdown



Key Findings

1. **Systematic Non-Compliance (76%)**
   The legal mandate fails in three-quarters of deliveries:

   - Never (30%): 15 respondents have NEVER witnessed mandated checks. These users don't even know what proper delivery should look like

- Sometimes (38%): 19 respondents experience random compliance, never knowing when safety protocols will be followed

- Never paid attention (8%): 4 respondents so disconnected they don't notice—indicating complete service invisibility

2. **The Compliance Lottery (38%)**
   The "sometimes" category is particularly problematic. Inconsistent service creates:

   - Uncertainty about when to insist on checks
   - Confusion about what's actually required
   - Gradual acceptance of non-compliance as normal

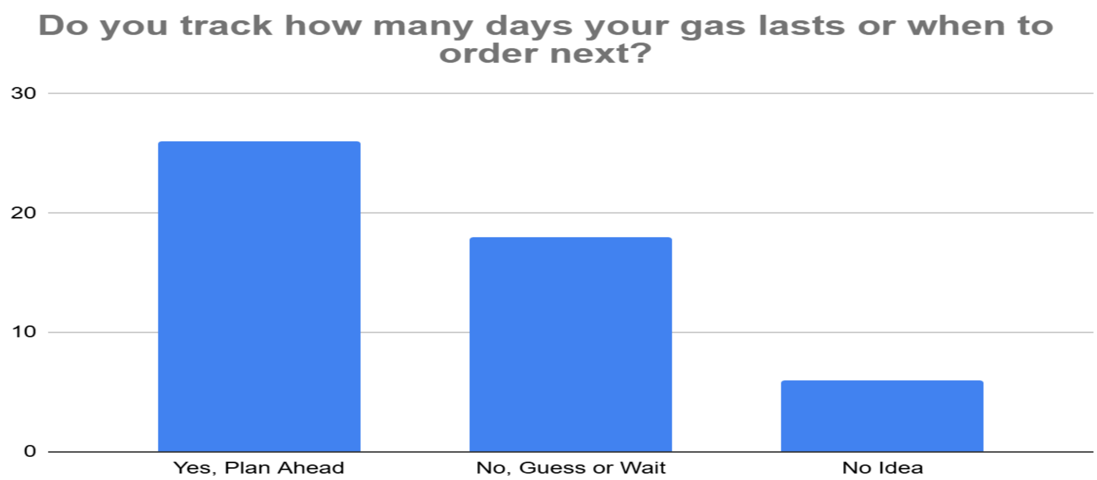
3. **The Gold Standard Minority (24%)**
   Only 12 respondents consistently receive legal safety checks. Geographic analysis shows these cluster in specific delivery routes, suggesting individual delivery personnel integrity rather than systemic compliance.

Critical Insight

When 76% of deliveries skip mandatory safety protocols, the "last line of defense" against faulty cylinders disappears. Users attempting checks themselves (67% try) lack both tools and training, creating false security.

"The delivery boy just drops the cylinder and extends his hand for money. When I asked about checking, he said 'Madam, you check if you want, I have 30 more deliveries." **- Homemaker, Age 42**

The Planning Impossibility



Key Findings

1. **Planning Attempt vs. Reality (52% vs. 78%)**
   26 respondents claim to "plan ahead," yet 39 experience exhaustion. This 26% delta reveals planning methods that fundamentally don't work:

   - Mental calculations based on "last month we used..."
   - Calendar reminders set arbitrarily ("order every 45 days")
   - Waiting for cylinder to "feel light"

2. **The Honest Guessers (36%)**
   18 respondents admit they "guess or wait", a more accurate assessment of their capability. Without consumption data, planning is literally guesswork.

3. **Complete Disengagement (12%)**
   6 respondents have "no idea" about tracking; they've given up attempting management, accepting chaos as inevitable.

Critical Insight

The system forces users into an impossible position: society expects responsible planning, but provides no tools for visibility. The 52% who "plan" are performing planning theater, going through motions without actual capability.

Data Summary

These data points triangulate to reveal three fundamental service failures:

- ❖ The Visibility Void: 78% exhaustion rate proves current monitoring methods (physical checking, guessing) are structurally inadequate

- ❖ The Safety Illusion: 72% suspect leaks but only 14% can verify, creating population living with unconfirmable fear

- ❖ The Compliance Collapse: 76% non-compliance with safety checks removes the only professional verification in the system

These aren't separate issues. They compound into a service ecosystem that actively generates anxiety rather than providing essential utility with confidence.

The research exposed alarming deficits in safety confidence and preparedness:

Gas Leak Detection Confidence Levels:

- Very confident: 0%
- Somewhat confident: 52%
- Not confident: 31%
- Unsure: 17%

The complete absence of high confidence represents systemic failure in safety assurance. Despite gas leaks being potentially fatal, not a single participant felt truly confident in their detection abilities. This finding gains severity when considered alongside the fact that 69% of respondents reported suspecting a gas leak at some point, yet only 19% knew how to perform basic leak detection using soapy water.

Safety Check Compliance:

Legal mandates require delivery personnel to verify cylinder weight and check for leaks in the customer's presence. However, research revealed:

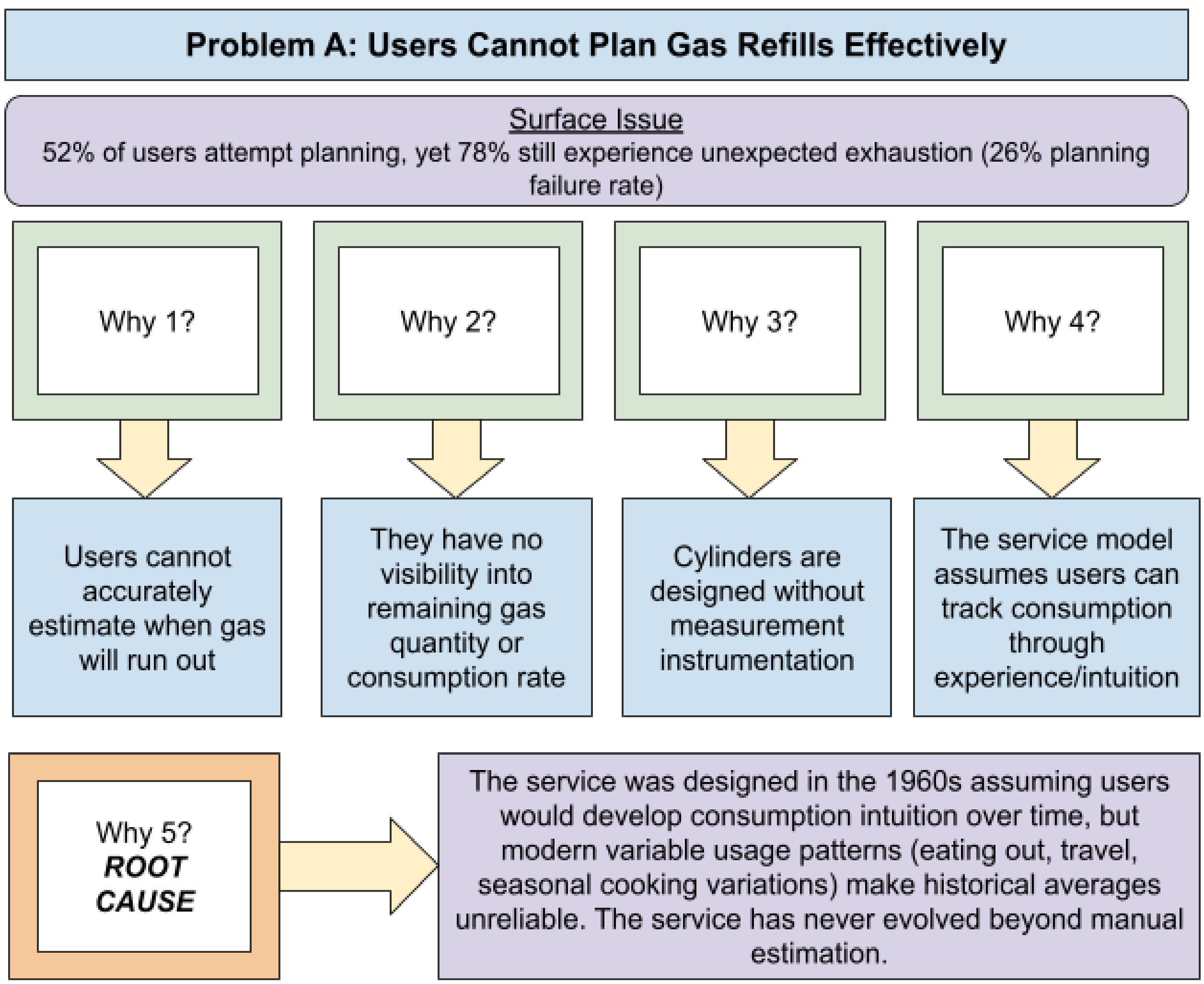
- Always performed: 10%
- Sometimes performed: 33%
- Rarely performed: 19%
- Never performed: 38%

This 90% non-compliance rate removes a critical safety checkpoint from the service chain, transferring risk entirely to unprepared consumers.

Root Cause Analysis

Urban Indian LPG management operates as a fundamentally broken service ecosystem where the delivery of an essential household utility has devolved into continuous crisis management. This is not merely a product deficiency or user education gap; it represents a systemic service design failure spanning multiple stakeholder touchpoints, service delivery protocols, and information architectures.

Applying the **5 Whys Methodology** to three critical service failures:



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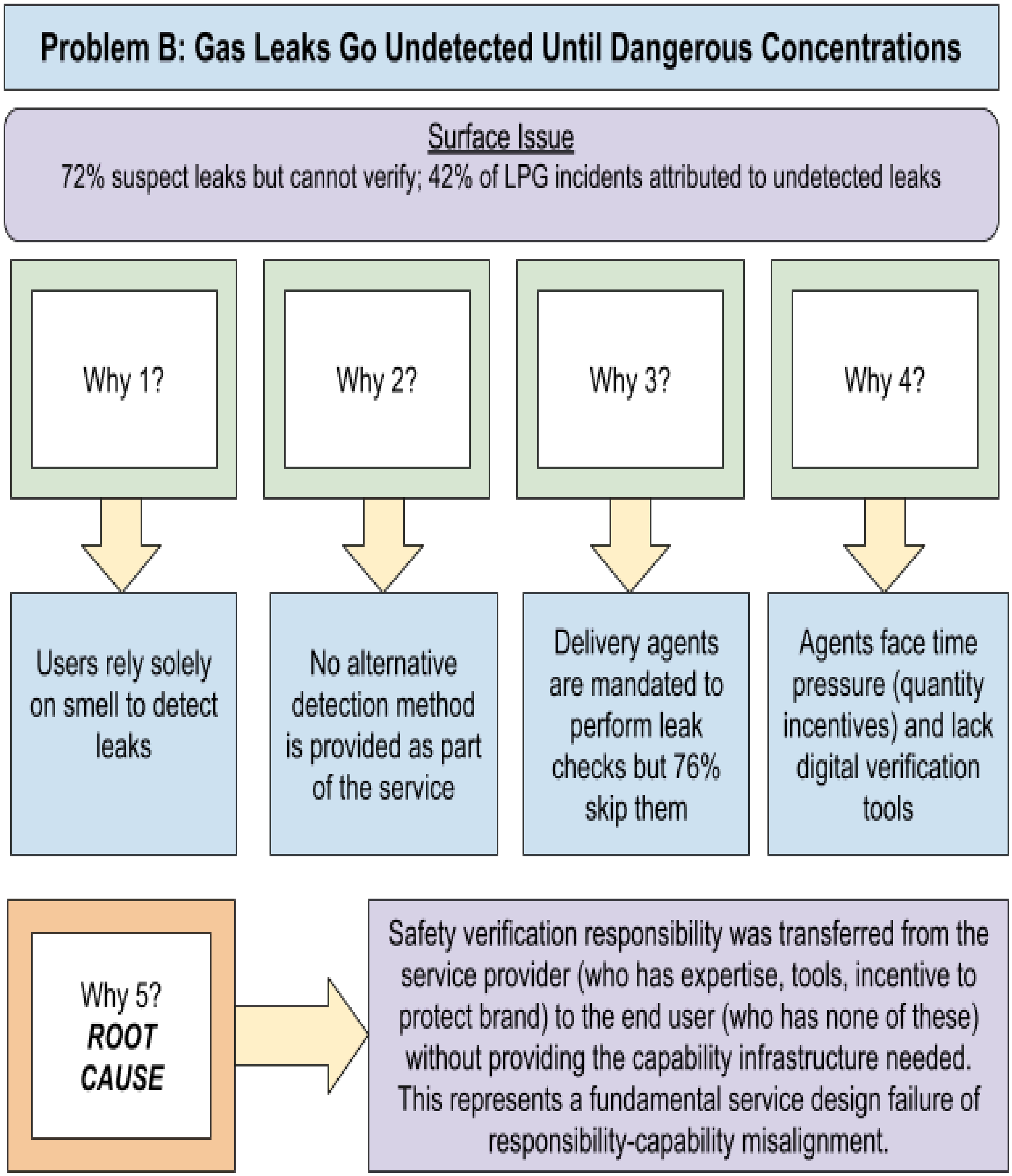

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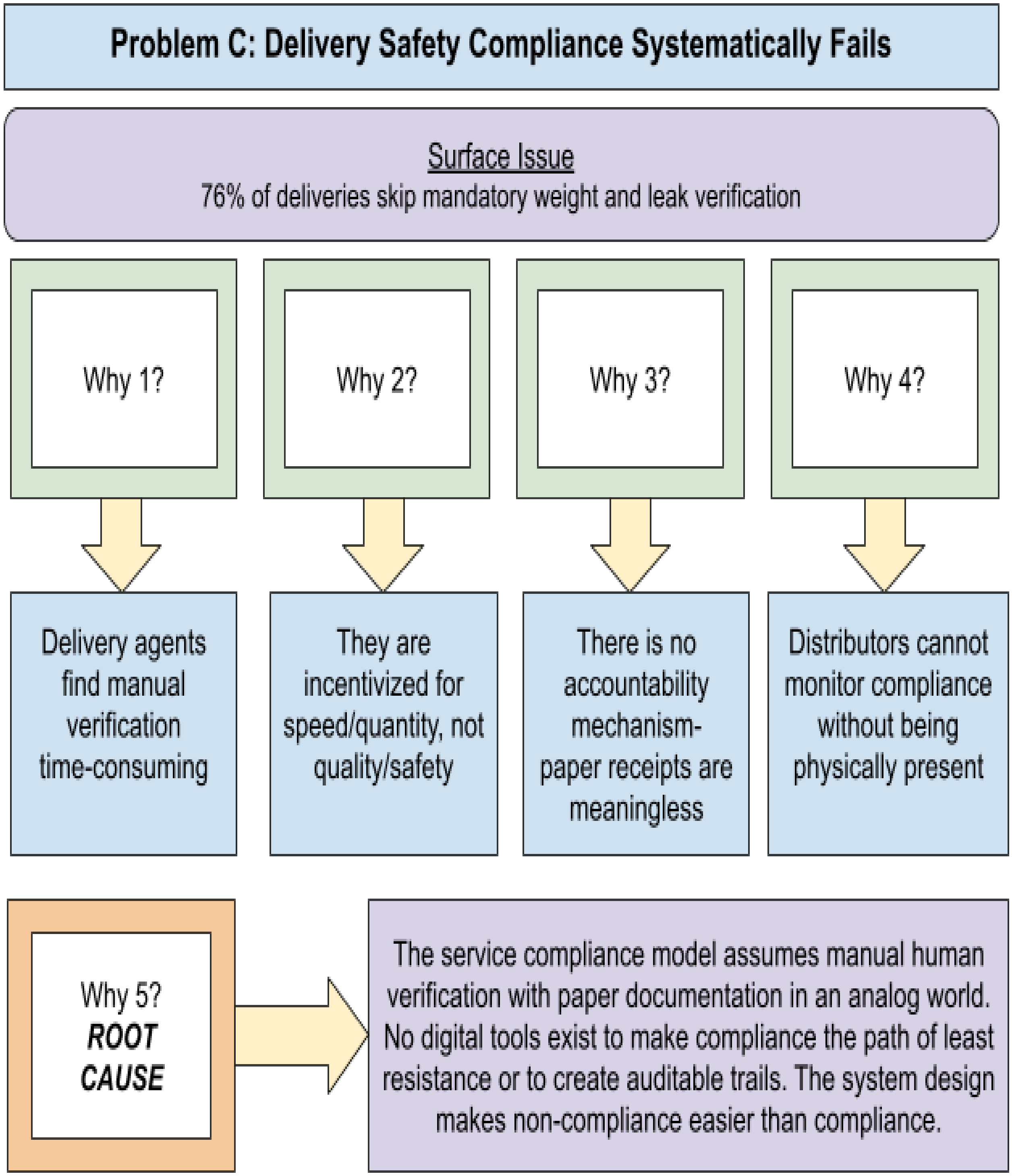

(Made With Google Drawings)

SafeCyl Affinity Map: Pain Point Clustering from User Research

200+ Individual Pain Points Synthesized into 5 Thematic Clusters (n=50 survey + 3 interviews)

VISIBILITY VOID
CRITICAL PRIORITY

No real-time gas level data
"58% just wait until it finishes"
Frequency: Daily

Physical lifting ±2-3kg error
"30% shake/lift cylinders—inaccurate"
Elderly: painful

Cannot estimate days remaining
"How much is left? 5 days? 15 days?"
Creates paralysis

No consumption pattern data
"Varies 35-50 days—no way to predict"
Planning impossible

Flame changes too late warning
"By time flame weak, only 2-4 days left"
Too little, too late

Complete info darkness
"Black box—cannot see inside at all"
Root cause

IMPACT: 88% lack reliable monitoring

SAFETY BLINDSPOT
CRITICAL PRIORITY

Unverifiable leak fears
"Smell something, but is it dangerous? Never know for sure"

3,068 incidents in 2023
"42% from undetected leaks" - PNGRB data
National crisis

Micro-leaks go unnoticed
"Below odor threshold but wasteful + risky"
Invisible danger

Elderly reduced smell sensitivity
"Age-related olfactory decline compounds vulnerability"

0% very confident in leak detection
"Despite fatal stakes, zero high confidence"
Systemic failure

Hypervigilance false alarms
"Every smell triggers panic, then dismissal"
Desensitization risk

ROOT: Responsibility without capability transfer

Visibility void enables safety blindspot

PLANNING IMPOSSIBILITY
HIGH PRIORITY

52% plan but 78% still exhaust
"26% planning failure rate—effort without efficacy"

Mental math fails pattern matching
"Last month 40 days, so this month... 45? 35? No idea."

Calendar reminders arbitrary dates
"Set Day 30 reminder but still uncertain: order now or wait?"

Consumption variability 20-30%
"Seasons, guests, travel, meal complexity invalidate averages"

Decision paralysis for 3-5 days
"Open app, stare at confirmation screen, close without action"

36% admit pure guesswork
"Honest assessment: I'm just guessing and hoping"

ROOT: 1960s service assumes intuition development

Safety blindspot creates planning crisis

ECONOMIC LOSS
HIGH PRIORITY

Mid-cooking exhaustion costs
"₹200-300 wasted food + ₹400-600 emergency restaurant per event"

Emergency booking premiums
"₹50-100 extra for rush delivery when ordered too late"

Backup cylinder opportunity cost
"₹1,500 deposit locked unnecessarily—could be in savings/investment"

Time loss in crisis management
"45-60 min per exhaustion event—calling restaurants, etc."

₹500-800 per household annually
"Preventable losses = ₹5,000 crore nationally"

Family tension & blame dynamics
"Birthday dinner for 12 ruined—we don't talk about it"

ROOT: System expects planning without data tools

Planning failure generates economic loss

COMPLIANCE COLLAPSE
MEDIUM PRIORITY

76% deliveries skip safety checks
"30% NEVER see checks 38% 'sometimes'—compliance lottery"

Agent time pressure (30-40/day)
"Verification adds 5-8 min—reduces daily deliveries & income"

Paper receipts meaningless
"User signs accepting quality they couldn't verify themselves"

User powerlessness to demand checks
"Asked agent to check, he said 'I have 30 more deliveries'—I backed down"

No digital audit trail for quality
"Cannot prove whether verification occurred during delivery"

Perverse incentives for agents
"Paid for quantity not quality—skipping checks directly increases pay"

ROOT: Analog verification in digital era

SYNTHESIS: These Five Clusters Compound into Service Ecosystem Failure

Key Insights from Affinity Clustering:

1. **Visibility is foundational:** Without real-time data, all other problems cascade (cannot plan, cannot verify safety, economic losses inevitable)
2. **Safety is non-negotiable:** Unlike economic/convenience pain points, safety failures have catastrophic consequences (3,068 incidents, ₹150Cr+ compensation)
3. **Compliance theater undermines trust:** Users lose faith in entire service when mandated protections fail 76% of time—creates systemic cynicism
4. **Economic costs are symptoms, not causes:** ₹5,000 crore national loss results from information architecture failure, not user negligence

Affinity mapping process: 200+ individual pain points from research → grouped by theme → 5 clusters emerged → validated against quantitative data → root causes identified

Critical Priority (Red) - Immediate intervention required

High Priority (Orange) - Significant user impact

Medium Priority (Yellow) - Systemic but addressable

Research Foundation: n=50 quantitative survey | n=3 qualitative interviews | 200+ pain points identified | 5 thematic clusters synthesized | 4 root causes isolated

Design Implication: SafeCyl must address all 5 clusters simultaneously—solving visibility alone is insufficient; service requires ecosystem transformation

Refined Problem Statement

The convergent analysis of behavioral patterns, survey results and the 5 Why’s Analysis highlight three main issues:

1. Information Asymmetry by Design
The 1960s service model never evolved to provide the consumption visibility that modern variable usage patterns demand,

2. Responsibility-Capability Misalignment
Safety verification was transferred from service providers (who possess expertise, tools, and brand protection incentives) to end users (who possess none of these) without providing detection infrastructure, and

3. Analog Compliance in a Digital Era
Paper-based verification systems make non-compliance easier than compliance, creating systematic protocol failures despite regulatory mandates.

These findings crystallize into our refined problem statement:

"How might we redesign the LPG service ecosystem to provide real-time consumption visibility, verifiable safety assurance, and proactive refill planning—transforming the 'responsibility-without-capability' model into an integrated service where monitoring, detection, and compliance are automated?"

This problem statement establishes the foundation for comprehensive service redesign spanning user-facing touchpoints (device, app, alerts), frontstage delivery interactions (digital verification, agent tools), backstage operations (distributor dashboards, predictive analytics), and support infrastructure (cloud systems, compliance reporting, ecosystem integration); addressing not merely the symptoms users experience but the systemic causes that generate those symptoms across 32.83 crore Indian households.

Current State Service Blueprint

Before designing interventions that transform the LPG service ecosystem, we must first comprehensively map and understand how the current service actually operates—not how it is supposed to work according to regulatory mandates or stakeholder intentions, but how it functions in ground reality across millions of daily user interactions. Service design methodology emphasizes that effective redesign requires deep understanding of existing touchpoints, backstage processes, information flows, and failure points; without this baseline mapping, proposed solutions risk addressing symptoms while missing systemic root causes or creating unintended consequences that ripple across the interconnected ecosystem.

LPG Cylinder Management Service - Present Situation

Detailed Explanation

STAGE 1: AWARENESS

User realizes they need to think about gas levels/safety

PHYSICAL EVIDENCE

1. LPG Cylinder (The Black Box) The 14.2kg commercial cylinder is a sealed metal container with no visual indicators of gas levels. Brand colors (red/blue/orange) and static information (tare weight, manufacturing date) provide no real-time operational data. The cylinder sits on kitchen floor connected via rubber hose to regulator—users interact with a complete "black box" where critical information (remaining gas) is entirely hidden. No gauge, no markings, no status indication exists.

2. Kitchen Environment & Spatial Constraints Urban metro kitchens (60-80 sq ft) create cramped quarters where cylinder checking requires clearing space and awkward bending. Surrounded by appliances and storage, the cylinder's inaccessibility compounds the difficulty of physical estimation methods, particularly for elderly users.

3. Regulator Connection System The brass regulator atop the cylinder represents the only user interface point. Connected via rubber hose, it produces faint hissing sounds users try to interpret as "normal" vs. "leak." The rubber seal cannot be visually verified. Users obsessively check the knob position (42% report checking before bed), yet it provides no information about gas quantity—only anxiety about whether it's properly closed.

4. Unreliable Sensory Cues Users depend on primitive detection: looking for frost (rare in Indian climate), listening for hissing amid kitchen noise, and sniffing for mercaptan odor. Kitchens retain post-cooking gas smell, and noses habituate to constant low-level odors—users cannot distinguish normal residual smell from dangerous slow leaks, creating false security.

5. Scattered Tracking Artifacts Users create makeshift systems: handwritten calendar notes, smartphone notes ("Last cylinder: 12-Oct"), or searching WhatsApp history for confirmation SMS (often deleted). These scattered records create retrieval friction. Memory-based tracking produces conflicting family accounts with no objective data.

CUSTOMER ACTIONS (Above Line of Interaction)

1. Physical Estimation Attempts (30% of users)
Users approach the cylinder, clear surrounding space, bend (risking back strain for elderly), and lift to "feel" the weight. A full cylinder weighs ~29kg; half-empty ~21-22kg. They attempt translating sensation to percentage: "Does this feel 50% or 30%?"

The Inaccuracy Problem: Human weight perception only distinguishes 15-20% differences reliably. The critical middle range (30-70% where booking decisions occur) is indistinguishable. User perception varies daily based on fatigue and grip strength—the same cylinder "feels different" day-to-day due to user state changes, not actual content changes. This creates ±2-3kg estimation error, making the method useless for precise timing.

Physical Toll: Elderly users (12% of sample) face genuine danger—interview participant Ramesh (65) described back pain lasting days after checking. Arthritic users find metal rim painful; reduced grip strength creates drop fears. Many abandon this method, defaulting to "wait until it finishes."

Shaking Variation: Some shake side-to-side listening for LPG sloshing, but acoustic properties in sealed containers aren't intuitive—70% full sounds similar to 40% full. This creates illusion of "doing something" without meaningful data.

Frequency Escalation: Early cycle (Days 1-10): checking every few days. Mid-cycle (Days 11-25): daily. Late-cycle (Days 26-35+): multiple times daily, particularly before cooking. This anxiety-driven behavior consumes mental energy and creates constant stress.

2. Olfactory Safety Monitoring (Continuous)
Users periodically sniff near cylinder/stove checking for mercaptan odor. Safety-anxious users report sniffing "every kitchen entry," sometimes 10-15 times daily.

Detection Delay: Human olfactory detection requires 1.5-2% LPG concentration taking 3-5 minutes to register consciously. Explosive range begins at 2.1%—essentially zero safety margin. By the time smell is detected, concentration approaches dangerous levels.

Habituation Problem: Kitchens retain residual cooking smells, and noses adapt to constant odors through neural habituation. Users living with faint background gas smell may not consciously register when genuine slow leaks elevate concentration. Users report "I thought I smelled gas, but it's always there a little, so I ignored it"—dangerous desensitization.

Anxiety-Hypervigilance Cycle: Some develop hypervigilance interpreting any kitchen odor as potential leak—cooking oil, burning food, garbage. False alarms train users to discount perceptions ("probably paranoid again"), creating risk when real leaks occur. The 72% who "suspected leaks" but cannot verify live in chronic unconfirmable concern.

3. Memory-Based Consumption Estimation

Users attempt mental math: "Last delivery was... 30 days ago? If last cylinder lasted 40 days, we have 10 days left." This fails because: (1) users cannot accurately recall dates without checking scattered records, (2) consumption varies significantly with seasons, guests, travel, meal complexity.

Pattern Recognition Failure: Users believe they have stable patterns ("usually 45 days per cylinder"), but actual data shows 20-30% variability. Households use cylinders in 35 days one cycle, 50 days the next. Users try predicting a complex variable system (daily consumption × days × seasons × occupancy) using gut feel and inaccurate memory.

Calendar Reminder Theater: The 52% who "plan ahead" set reminders: "Check gas - Day 30." These arbitrary dates aren't based on actual consumption—they're hopeful guesses. Reminders don't reduce uncertainty; they schedule the moment of uncertainty. This is "planning theater"—performing planning behaviors without data infrastructure that enables actual planning.

4. Flame Observation (Indirect Indicator)

Users watch for flame changes: healthy flame is blue with consistent height; depleted cylinders produce smaller, yellower flames taking longer to cook. Users notice during cooking: "Why is this taking so long? Is the flame weak?"

Ambiguity Problem: Flame changes can result from: low gas, burner clogging, regulator malfunction, or poor ventilation. Users cannot distinguish causes—some clean burners when gas is actually low; others assume dirty burner when gas is critically low.

Late-Stage Warning: By the time flames change noticeably, cylinders are 10-20% full—only 2-4 days remaining. This late warning compresses booking window, often overlapping with 2-7 day delivery time, creating high exhaustion risk.

5. Family Consultation (Distributed Uncertainty)

Users ask household members: "Did you check the gas?" "Should we order?" "When did we get it last?" This distributes decision burden but doesn't resolve information void—multiple people without data collectively have no data. Responses vary ("three weeks ago" vs. "at least a month"), creating confusion.

Responsibility Diffusion: When everyone monitors but no one has tools, responsibility diffuses. Research shows 6% delegate entirely to another household member; 6% attempt estimation based on others' reports. This creates accountability gaps—everyone assumes someone else is tracking.

6. Anxiety Behaviors

Pre-Sleep Checking: Safety-anxious users (42%) check regulator position nightly before bed. This becomes compulsive ritual—users report "I can't sleep unless I check." Checking provides temporary relief but doesn't address root concern.

Backup Cylinder Maintenance: Higher-income households cope by maintaining backup cylinders requiring ₹1,500 deposit (opportunity cost) and scarce storage space. This provides psychological safety but represents economic inefficiency—capital locked in inventory that proper visibility would eliminate.

Consumption Reduction: As perceived depletion progresses, users ration: simpler meals, electric kettle instead of gas for boiling, eating out more. These behaviors disrupt routines and create culinary limitations—users pre-emptively compromising lives due to information void.

FRONT-OF-STAGE INTERACTIONS (Below Line of Interaction)

[NO EMPLOYEE/SYSTEM INTERACTION AT AWARENESS STAGE]

Users struggle alone with primitive estimation, anxiety, and uncertainty. No proactive outreach from distributors—no calls, no app notifications, no alerts. Distributors possess historical data (last delivery date, average days between orders) but information remains locked in databases, unexposed to users.

Service Provider Silence: The party with least information (user) drives the process, while the party with most information (distributor/OMC) remains passive. Compare to electricity meters (real-time consumption), mobile data plans (GB remaining apps), water meters (monthly usage bills)—LPG alone operates in information darkness, generating continuous anxiety by design.

BACK-OF-STAGE INTERACTIONS (Below Line of Visibility)

[NO BACKEND ACTIVITY DURING AWARENESS]

Distributors and OMCs have no active monitoring while users struggle. Backend remains idle, waiting for user-initiated booking to trigger activity.

SUPPORT PROCESSES *(Infrastructure Layer)*

1. Distributor Database (Passive Storage) Contains consumer numbers, addresses, delivery history, average ordering frequency (e.g., "orders every 38 days"), payment patterns.

Critical Failure: Data exists but isn't used proactively. Historical patterns could enable predictive outreach ("likely needs gas in 3-5 days") but distributors lack incentive/infrastructure. They operate reactively—waiting for orders, then fulfilling. Data sits dormant.

2. OMC Central Systems (Minimal Engagement) Track national distribution, regional inventory, distributor performance—but NOT individual consumption patterns, predicted household demand, or safety incidents correlated to compliance.

Missed Opportunity: OMCs could develop apps/SMS services providing visibility, but monopolistic market structure reduces competitive pressure for innovation. Users are captive; switching is difficult.

3. Regulatory Framework (Unenforced) PNGRB mandates and BIS standards govern specifications, but no monitoring of whether consumers have adequate management tools. Regulations assume intuitive consumption understanding—a 1960s assumption unchanged despite technology advances.

Failure Points:

- Complete information asymmetry (users: zero data; providers: historical data unused)
- No early warning infrastructure
- Anxiety generation by design
- Economic inefficiency (backup cylinders, consumption rationing)
- Safety risk (olfactory detection delays, habituation)

STAGE 2: CONSIDERATION

User actively thinks about booking but hasn't committed

PHYSICAL EVIDENCE

1. Intensified Physical Checking Cylinders lifted/shaken multiple times daily vs. every few days. Kitchen floor shows scuff marks from repeated repositioning. Users' hands show redness from metal rim—evidence of frequency/force. Family members comment: "Why do you keep checking?"

2. Digital Artifacts Foreground

- **Booking app** moves from buried folder to home screen—visual reminder creating ambient anxiety
- **Browser tabs** remain open for days (booking page visited but not completed)—indicating intention without commitment
- **SMS history** searched extensively for last confirmation (5-10 min scrolling), some screenshot and save to Photos as workaround

3. Behavioral Changes

- **Hesitant cooking:** Simple meals preferred ("let's not make biryani that needs 2 hours—gas might run out")
- **Restaurant app browsing:** Users prepare contingencies, sometimes adding items to cart without ordering
- **Family discussions:** Dinner conversations include gas status without reaching consensus (no one has objective data)

CUSTOMER ACTIONS *(Above Line of Interaction)*

1. Booking App/Website Navigation (Multi-Step Friction)

Authentication Hassle: Users don't remember credentials created during initial registration. Password reset flows trigger: Enter mobile → Receive OTP → Create password → Re-enter. This adds 3-5 minutes; 15-20% abandon here.

Consumer Number Retrieval: Must enter 17-digit code printed on distributor card. Users physically locate card (document drawer, kitchen cabinet) or search phone for photos, sometimes WhatsApp family members: "Do you have photo of gas card?" Adds 2-5 minutes.

Form Completion: Confirm cylinder type, delivery address, contact number, payment method. Each field requires attention, sometimes correction. For elderly/low-digital-literacy users, each dropdown represents cognitive load.

Delivery Timing Ambiguity: Interface shows "2-3 days" or "within 7 days" without specific slots. Some offer broad windows (Morning/Evening), many provide no choice. Users cannot plan around delivery because delivery cannot be planned.

Payment Decision: 70% prefer COD as psychological leverage: "If cylinder seems wrong, I can refuse payment." Prepayment feels risky (what if faulty/underweight?). COD requires exact change (₹850-1,050), adding mental overhead.

Final Hesitation: At confirmation screen, users pause: "Is it really time? Too early and they might refuse; too late and gas runs out during 2-7 day wait?" This crystallizes the information void—a straightforward decision becomes high-stakes guessing.

Total Friction: 6-12 minutes (tech-comfortable) to 15-25 minutes (low-literacy). Compare to food delivery (2-3 min) or ride-hailing (1-2 min)—LPG friction is 3-10x higher, creating abandonment/procrastination.

2. Direct Phone Calls (Traditional Channel)

Users call seeking reassurance ("Is it time to order?") or clarity ("When exactly will you deliver?"). Call may ring unanswered (distributor busy managing warehouse). If answered, user gets generic response: "Delivery in 2-3 days. Please book through app."

Information Asymmetry Continues: Agents have no access to consumption history enabling helpful personalized responses. Call adds human touch but no intelligence. Users hoped for guidance; agents operate in same information void.

3. Extended Decision Paralysis

- **Repeated app opening:** Navigate to confirmation, then close without submitting—multiple times daily seeking certainty that doesn't emerge
- **Peer consultation:** Ask neighbors "How do you know when to order?" Responses provide comfort (shared uncertainty) but no data
- **Mental scenarios:** Elaborate simulations: "If I order Tuesday, comes Thursday, but gas runs out Wednesday..." Complex probabilistic reasoning without sufficient data creates exhausting cognitive load

4. Backup Planning

- **Restaurant pre-selection:** Identify quick-delivery options, operating hours, menu—10-15 minutes of contingency planning
- **Family warnings:** "Be careful with gas, we're running low"—distributes anxiety, creates conservation pressure, pre-assigns blame if exhaustion occurs

5. Physical Checking Escalation Frequency increases to 3-4 times daily: morning, midday, evening, pre-sleep. Each check 2-3 minutes; aggregated: 8-12 minutes daily on futile checking. Users compare sensations across checks, but human perception varies enough to add noise rather than signal.

FRONT-OF-STAGE INTERACTIONS (Below Line of Interaction)

1. Booking Confirmation (If Completed) Automated SMS/Email: "Order `#ABC123` confirmed for Consumer #XXXX. Expected delivery: 2-3 days."

Message Inadequacy: No specific date, time slot, tracking number. "2-3 days" is vague—users must remain available during business hours for multiple days. Users screenshot message, attempt replying (no-reply number), reinforcing one-way communication.

2. Call Center Interaction (If User Calls)

Agent's Limited Capability: Access to basic systems (order placement, consumer database, inventory yes/no) but NOT consumption history, precise scheduling, real-time tracking, past feedback.

Scripted Conversation:

- Agent: "How may I help?"
- User: "When will delivery arrive?"
- Agent: "Within 2-3 days."
- User: "But WHEN exactly? I need to be home."
- Agent: "Agent will call 30 minutes before."
- User: "But which DAY?"
- Agent: "Within 2-3 days."

User Frustration: Users seek specificity, agents provide generality. Not agent's fault—backend lacks sophistication for precise scheduling. Agents handle 50-80 calls/day with 2-3 minute targets, creating time pressure manifesting as curt responses users interpret as rudeness.

3. No Proactive Outreach (Service Gap)

What DOESN'T happen: No proactive SMS ("Based on your 40-day consumption, order in 3-5 days"), no app notification ("32 days since last delivery"), no email with insights ("You use 0.35kg daily, likely 15% remaining").

Electricity meters show real-time consumption; mobile apps alert at 50%, 80%, 90%. LPG alone operates without basic utility intelligence, forcing guessing games.

BACK-OF-STAGE INTERACTIONS (Below Line of Visibility)

1. Order Processing Booking enters distributor queue creating database entry: consumer number, cylinder type, timestamp, payment method, status: "Pending." Orders don't trigger immediate scheduling—they accumulate for batch processing (once/twice daily). 9 AM Monday and 11 PM Monday orders treated identically despite 14-hour difference.

2. Manual Route Planning Each morning, distributor staff review accumulated orders and perform route planning—often manual human judgment: "8 orders in Sector 15, 6 in Sector 22—Agent Raju takes both." This is reactive order grouping, not predictive demand planning.

Process Inefficiencies: Orders arrive unpredictably (emergency bookings create spikes); route planning is reactive, not optimized; delivery timing depends on how many other orders in same area (network effects user cannot see).

SUPPORT PROCESSES

OMC Booking System: Routes orders to correct distributor based on consumer number, generates order ID, logs timestamp. No predictive demand forecasting—just processes incoming requests.

Distributor Local System: Receives order, updates inventory (-1 available cylinder), adds to delivery queue. Excel-based or basic software—not integrated with smart logistics.

Payment Gateway: Processes prepayment if selected. Most users avoid due to "what if cylinder is faulty?" fear, preferring COD for negotiation leverage.

Systemic Issues:

- Demand unpredictability (emergency bookings create chaos)
- No load balancing (10 orders one day, 50 the next)
- Inventory risk (distributors maintain buffer stock unable to forecast)

STAGE 3: DECISION

User commits to booking or continues delaying

PHYSICAL EVIDENCE

1. Booking Confirmation SMS (If Booked) "Your order confirmed. Delivery in 2-3 days." Users screenshot for proof and reminder to "wait anxiously for next 2-7 days."

2. Behavioral Adjustments (If NOT Booked)

- **Reduced cooking plans:** Simple meals, avoid long cooking
- **Increased checking:** Physical lifting 3-4x daily
- **Backup activated (wealth-dependent):** Peace of mind at ₹1,500 deposit cost

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CUSTOMER ACTIONS (Above Line of Interaction)

1. Booking Completion (6-8 steps/taps) Enter consumer number (search phone for it) → Select cylinder type → Confirm address → Choose payment (COD preferred) → Receive OTP → Enter OTP → Confirm. **Total time: 3-5 minutes, creates post-booking anxiety:** "Did they receive it? When exactly will it come?"

2. OR Decision to Delay Set reminder "3 days later," inform family "careful with gas usage," anxiety increases daily until booking finally made. Research shows 52% "plan ahead" but 78% still experience exhaustion—26% who delay too long.

3. Tracking Void No real-time delivery tracking (unlike food delivery apps). Users receive vague confirmation, then wait in information darkness for "2-7 days." Cannot schedule activities confidently—must remain available during business hours for multiple days.

FRONT-OF-STAGE INTERACTIONS

Booking Confirmation (Automated): Message provides no specific date, time, tracking number, or contact info for queries. Vague timeline generates continued uncertainty.

[NO HUMAN INTERACTION UNLESS USER CALLS] Some call distributor: "I just booked online, did you get it?" Distributor confirms: "Yes, we'll deliver soon." "Soon" = continued ambiguity.

BACK-OF-STAGE INTERACTIONS

Order Processing: Booking enters distributor queue. Manual route planning by distributor staff groups orders by locality, assigns to agents based on territory. **No optimization algorithm**—done by experience/"feel." Delivery scheduled "sometime in next 2-7 days," no precise slot due to unpredictable demand spikes.

Inventory Reserved: One cylinder marked for user in warehouse ledger.

Payment Status: COD flagged; agent will collect on delivery.

SUPPORT PROCESSES

OMC Central System: Routes order to distributor, generates ID, logs timestamp. No predictive demand forecasting.

Distributor Local System: Receives notification, updates inventory, adds to queue. Excel-based or basic software—not integrated with smart logistics.

Systemic Issues:

- Demand unpredictability creates chaos
- No load balancing
- Inventory risk (distributors cannot forecast)

The system implements a sophisticated multi-layer connectivity framework that ensures reliable data flow from sensors to end users while maintaining safety-critical operations even during network disruptions.

STAGE 4: SERVICE (Delivery)

Cylinder is delivered to household

PHYSICAL EVIDENCE

1. Delivery Agent Arrival The delivery agent arrives on motorcycle or small tempo (Tata Ace) carrying multiple cylinders—typically 8-12 strapped together in the vehicle. Agent's appearance varies: some wear branded distributor uniforms (minority), most wear casual clothes with no identification. The vehicle itself is often unmarked or has basic distributor name painted on side. Users hear the doorbell or phone call: "LPG delivery for [name/address]."

2. The New Cylinder (Visually Identical) The delivered cylinder looks exactly like the previous one—same brand, same size, same metal exterior. Users have no visual method to verify quality, weight accuracy, or safety. The cylinder is a commodity object where individual units are indistinguishable without instrumentation. The painted exterior might show minor dents or scratches from warehouse handling, but these provide no information about internal gas content or valve integrity.

3. Mandated Equipment (Usually Absent) Legal requirements specify delivery agents should carry:

- **Weighing scale** (to verify 14.2kg gas weight in customer's presence)
- **Soap water solution** (to apply on connections and check for bubbles indicating leaks)

Research Reality: 76% of deliveries occur without these tools present. Users in our survey report:

- 30% have NEVER seen weighing equipment during any delivery
- 38% see it "sometimes" (inconsistent, unpredictable)
- Only 24% consistently receive proper verification

The absence of verification equipment becomes normalized—users don't even know to expect it. First-time LPG users have no reference for "proper delivery" and assume rushed, unchecked delivery is standard procedure.

4. Payment Exchange Artifacts

- **Cash (COD preferred by 70%):** Agent carries change in pocket/pouch, counts notes in doorway, provides handwritten/printed receipt
- **Digital payment:** Agent shows QR code (Paytm/PhonePe) on phone, user scans and confirms—takes extra 2 minutes, agents prefer cash for speed
- **Paper receipt/bill:** Handwritten carbon copy or thermal-printed slip, often illegible, immediately filed (or lost) by user

5. The Exchange Moment (60-90 seconds total) The entire delivery interaction—from doorbell to agent departure—averages 60-90 seconds. This rushed timeframe makes meaningful safety verification practically impossible even if mandated equipment were present. Users barely have time to direct agent to kitchen, watch cylinder placement, sign receipt, and make payment before agent is walking out the door.

CUSTOMER ACTIONS (Above Line of Interaction)

1. Receiving Delivery User hears doorbell/call, opens door, directs agent to kitchen. Watches agent carry new cylinder (often through living room, past furniture—creating minor household disruption), place it in corner where old cylinder sits. Agent disconnects regulator from old cylinder, connects to new one—sometimes checking for gas release sound (brief hissing when connection made indicates flow).

2. Attempted Verification (Minority Behavior) Aware users (those who know verification is legally mandated) attempt to request checks: "Can you please verify the weight?" or "Is there any leak test?"

Agent Responses (Research-Documented):

- "Madam, you check if you want, I have 30 more deliveries" (actual quote from research participant)
- "Everything is fine, don't worry" (dismissive reassurance)
- "The cylinder is filled at the plant, no problem" (deflecting responsibility)
- [Silence while extending hand for payment] (non-verbal pressure to conclude transaction)

User's Dilemma: Users face social pressure not to "cause trouble" or "delay the busy agent." They lack confidence/knowledge to insist forcefully. Power imbalance (agent is service provider representative, user is "just a customer") creates deference. Users back down, signing receipt without verification, internalizing "maybe I'm being too demanding."

3. Receipt Signing (Habitual, Unread) Users sign paper receipt—often without reading the fine print stating "customer acknowledges receipt of properly filled, leak-tested cylinder." This signature legally transfers liability from distributor to user despite no actual verification occurring. Users sign habitually, treating it as formality rather than legal acknowledgment of quality acceptance.

4. Payment Transaction

COD (Majority Method): User hands cash (₹850-1,050 depending on location/subsidies), agent counts and provides change. Transaction happens in doorway, often with agent holding cylinder in one hand, accepting money with other—body language communicating urgency to leave.

Digital Payment: User scans agent's QR code, confirms amount in payment app, shows "Payment Successful" screen to agent. Agent verifies on their phone, waits for confirmation notification (sometimes 10-20 sec delay causing awkward standing silence), then leaves. This method adds 2-3 minutes vs. cash, which agents dislike—they're paid/bonused by deliveries completed, not payment method convenience.

5. Post-Delivery Immediate Actions Agent removes old/empty cylinder, loads it onto vehicle. User closes door. Immediate action: turn on stove briefly to verify gas flows (testing the new connection). If flame appears, user feels temporary relief. If no flame or weak flame, panic sets in: "Did they connect it properly? Is there a problem?" requiring calling distributor/agent back (adds 15-30 minute stress).

The Verification Gap: User has NO way to verify: (1) Is this cylinder actually 14.2kg of gas, or short-filled? (2) Are there leaks in connections? (3) Is the regulator properly sealed? They must simply trust the agent performed mandated checks despite not witnessing them. This trust is unearned—76% non-compliance rate means trusting is statistically irrational.

6. Emotional Response (Relief Tinged with Anxiety) Users experience temporary relief ("We have gas now!") but underlying anxiety remains: "Did I order too early? Could we have gone another week?" (second-guessing wastes mental energy) and "Is there a leak I can't detect?" (safety anxiety transfers from old to new cylinder). The cycle doesn't end; it restarts at Day 1.

FRONT-OF-STAGE INTERACTIONS
(Below Line of Interaction, Above Line of Visibility)

1. Delivery Agent Visible Actions

Physical Labor: Agent carries new cylinder from vehicle to customer's kitchen—navigating stairs (in buildings without elevators, carrying 29kg up 2-3 floors), tight doorways, household obstacles. This physical exertion creates time pressure: "The faster I finish, the sooner I can rest."

Cylinder Exchange Process:

1. Places new cylinder where user indicates
2. Disconnects regulator from old cylinder (checking rubber hose condition visually—but not replacing even if worn unless customer specifically requests and pays extra)
3. Connects regulator to new cylinder (tightening with hand, sometimes brief wrench use)
4. Tests connection by opening regulator briefly, listening for gas release (2-second check)
5. Retrieves old/empty cylinder, carries back to vehicle
6. Returns to user's door, presents bill/receipt
7. Collects payment
8. Leaves

Total Time: 60-90 seconds from entry to exit.

[LEGALLY MANDATED BUT SKIPPED 76% OF THE TIME]:

What SHOULD Happen (Per PNGRB Regulations):

- **Weighing Verification:** Place cylinder on weighing scale, show customer the weight reading, confirm it matches gross weight specification (29kg for 14.2kg commercial)
- **Leak Testing:** Apply soapy water solution to all connection points (regulator-cylinder valve junction, hose-regulator junction, hose-stove junction), check for bubble formation indicating leaks
- **Safety Explanation:** Inform customer of proper regulator usage, ventilation requirements, emergency contact numbers
- **Previous Cylinder Query:** Ask "Any issues with the last cylinder?" to track recurring problems

Why Verification is Skipped—Agent's Perspective:

Time Pressure Economics: Agents are assigned 30-40 deliveries per day with expected completion within 8-hour shift. This allows approximately 12-16 minutes per delivery including travel time between locations. Actual verification (weighing: 3 min setup + execution; leak test: 2-3 min application + observation) would add 5-8 minutes per delivery, reducing possible daily deliveries from 35 to 20-25, directly cutting agent earnings (typically paid ₹15-25 per delivery or monthly salary with quantity-based bonuses).

Incentive Misalignment: Agents are rewarded for speed/quantity, not quality/safety. Performing mandated checks feels like penalty (reducing income) rather than duty. No quality bonuses exist; no customer satisfaction metrics affect agent compensation. The system structurally incentivizes non-compliance.

Equipment Burden: Carrying weighing scale (additional 5-8kg equipment) and soap solution (bottles that leak, require refilling) adds physical burden to already labor-intensive work (carrying 29kg cylinders up stairs repeatedly). Agents leave equipment in vehicle or at warehouse, rationalizing "customers don't ask for it anyway."

Enforcement Absence: No supervisor presence during deliveries; no digital audit trail creating accountability; no customer feedback immediately affecting agent performance review. Paper receipts (signed by customers) provide meaningless documentation—signatures are collected whether verification occurred or not.

Customer Doesn't Insist: 30% of users have NEVER seen verification, don't know it's legally required. 38% see it "sometimes," accepting inconsistency as normal. Only 24% consistently demand it, and even they face agent resistance ("I have 30 more deliveries"). Users feel powerless to enforce legal mandates.

2. Delivery Confirmation Logging (Often Delayed) Agent should mark order "delivered" in system real-time using smartphone app, but many fill logs at day's end (batch processing). Customer cannot track during delivery—sees "Out for Delivery" status that doesn't update until hours later when agent reconciles paperwork.

BACK-OF-STAGE INTERACTIONS *(Below Line of Visibility)*

1. Inventory Update Distributor system registers: -1 filled cylinder (inventory depleted), +1 empty returned (exchange stock). **No quality assurance checkpoint**—system assumes all delivered cylinders are compliant, properly filled, leak-free. No verification of agent's verification occurs.

2. Payment Reconciliation

- **COD:** Agent carries collected cash, reconciles with distributor at day's end (counting, matching against delivery receipts). Discrepancies create disputes: "I delivered 35 but only have cash for 33—did I forget two payments or did customers not pay?"
- **Digital:** Automatically logged in payment gateway, distributor receives settlement within 24-48 hours

Direct cash handling creates reconciliation complexity and potential discrepancy issues, but distributors prefer it for immediate liquidity.

3. Customer Database Update Consumer record updated: Last delivery date, payment received, empty cylinder returned. **No quality metrics logged:** Was verification performed? Customer satisfaction? Delivery timeliness? These dimensions go untracked—system cares only about transaction completion (yes/no), not service quality.

Process Gaps:

- No real-time tracking visible to customer (unlike food delivery with live GPS)
- No photo documentation of delivery (no proof cylinder was delivered to correct address with correct condition)
- No immediate feedback prompt post-delivery
- No quality assurance layer between agent and customer

SUPPORT PROCESSES

1. Distributor Dispatch System Assigns 30-40 deliveries/day to each agent. Routes planned manually or basic optimization (geographic clustering). **No buffer time for quality checks**—schedules assume 3-5 min per delivery including travel. If agent performs mandated verification (5-8 min), schedule breaks down by midday.

2. OMC Compliance Monitoring (Theoretical)

What Should Exist:

- Periodic audits of distributors

- Random customer feedback calls
- Complaint redressal system
- Compliance performance metrics tied to distributor contracts

What Actually Exists: 90% non-compliance rate indicates monitoring is ineffective or non-existent. Users report complaints go unanswered or result in "we'll look into it" with no follow-up. Regulatory framework lacks enforcement teeth.

3. Paper Documentation System Delivery receipts filed physically in distributor office. **No central digital repository** of delivery logs with verification status. Cannot data-mine for patterns: which agents skip checks, which areas have complaints, which customers need intervention.

The Auditability Failure: If accident occurs and investigation asks "Was this cylinder properly verified upon delivery?", the paper receipt shows customer signature (implying acceptance) but no actual evidence verification occurred. This protects distributor legally while transferring liability to user who couldn't verify.

Systemic Failures:

- **Perverse incentive alignment:** System rewards speed/quantity, penalizes thoroughness/quality
- **No accountability infrastructure:** Paper trails meaningless, no digital audit capability
- **Regulation without enforcement:** Laws exist but no mechanism ensures compliance
- **Trust-based system:** Entire safety checkpoint relies on trusting agent performed checks—user has no verification method

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STAGE 5: LOYALTY (Post-Delivery)

User's ongoing relationship after delivery

PHYSICAL EVIDENCE

1. New Cylinder in Kitchen User's primary interface for next 30-45 days. Cycle restarts: Day 1-10 relaxed ("gas is full"), Day 11-25 background awareness returns, Day 26-35 active worry, Day 36+ high anxiety. Same physical artifact, same information void.

2. Documentation Artifacts

- **Paper receipt:** Stored (or lost) in household documents
- **SMS history:** Often deleted to clear storage
- **Mental satisfaction/dissatisfaction:** No structured feedback mechanism

3. Coping Mechanism Evidence

- **Backup cylinder** (if maintained): Continues requiring storage space and locked deposit
- **Restaurant expense receipts:** If gas exhausted unexpectedly during wait period, emergency orders placed
- **Family tension artifacts:** Arguments about "who should have ordered earlier" create emotional residue

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CUSTOMER ACTIONS (Above Line of Interaction)

1. Normal Cooking Resumption User resumes regular meal preparation with new cylinder, but anxiety cycle doesn't end—it restarts. The psychological relief is temporary (Days 1-7), then background awareness returns (Days 8-20), escalating to active worry (Days 21-35+), culminating in crisis or just-in-time booking.

2. Word-of-Mouth Sharing (Informal Feedback)

Negative Experiences: Users tell neighbors/friends if delivery was particularly bad: "The agent didn't even check the weight—just dropped it and left!" These stories spread through WhatsApp groups and apartment complex interactions, building general service dissatisfaction but not reaching formal channels where distributors might respond.

Positive Experiences (Rare): Occasionally users experience proper verification: "For once, they actually did the leak test!" This surprise indicates how rare compliance is—it's notable rather than expected. Users share this as exceptional service, not standard practice.

The Feedback Dead-End: These informal conversations happen in social spaces—housing society WhatsApp groups, neighborhood gatherings, family dinners—but never reach OMCs or distributors. No formal complaint channels capture this sentiment. Users assume "complaining won't help" based on past experience of unaddressed complaints.

3. Loyalty Decision (Monopolistic Lock-In)

Reality: No Choice: LPG consumer numbers are tied to specific distributors based on address. Switching requires re-registration, new consumer number, potential loss of subsidy benefits, bureaucratic paperwork. Users are effectively captive customers regardless of service quality.

Learned Helplessness: Users develop "this is just how LPG works" mentality. Dissatisfaction doesn't translate to switching (impossible) or complaint (ineffective)—it translates to acceptance and personal coping mechanisms (backup cylinders, restaurant budgets, consumption rationing).

No Competitive Pressure: Distributors face minimal competitive threat. Unlike telecom (users switch carriers easily) or food delivery (multiple apps compete for loyalty), LPG operates as localized monopoly. This removes primary market force that typically drives service improvement.

4. Coping Mechanism Development

Backup Cylinder Strategy (Wealth-Dependent): Higher-income users (>₹60k monthly) continue maintaining backup cylinders despite new cylinder arrival. The backup provides "never run out" guarantee worth ₹1,500 opportunity cost to them. This coping mechanism addresses symptom (exhaustion risk) rather than cause (information void), but users feel it's their only option.

Consumption Adaptation: Users learn to cook simpler meals during late-cycle (Days 25-35+): "I avoid elaborate dishes when gas might be running low." This lifestyle compromise becomes normalized—users don't even consciously recognize they're modifying behavior around service inadequacy.

Restaurant Relationships: Frequent LPG exhaustion creates repeat emergency restaurant orders. Users develop go-to restaurants for crisis situations, sometimes mentioning to restaurant owners "we might call with large order suddenly when our gas runs out—can you handle it?" This informal service relationship (restaurant as LPG backup) highlights the service ecosystem's failure to be reliable.

Mental Health Impact: Chronic anxiety about gas levels and safety creates low-grade constant stress. Users describe "It's always there in the back of my mind, using mental energy." Interview participant Priya (35) reports difficulty sleeping some nights: "I lie awake thinking: did I close the regulator? Is there a leak? When should I order?" This psychological burden is unmeasured cost of service design failure.

5. No Service Improvement Actions

Users Don't Complain Formally Because:

- Previous complaints went unaddressed ("we'll look into it" with no follow-up)
- Complaint process is cumbersome (call distributor, explain situation, get ticket number, wait indefinitely)
- Monopolistic lock-in means complaining risks antagonizing only available service provider
- Learned helplessness: "Nothing will change, so why bother?"

The Missing Feedback Loop: In well-designed service ecosystems, customer feedback drives continuous improvement. Here, negative experiences evaporate into informal social complaints without reaching decision-makers who could act. Distributors and OMCs operate without real-time service quality data, making improvement impossible even if they wanted to.

FRONT-OF-STAGE INTERACTIONS (Below Line of Interaction)

1. Post-Delivery Silence (Minimal Engagement)

[NO PROACTIVE FOLLOW-UP OCCURS]:

- No "How was your delivery?" call/SMS from distributor
- No customer satisfaction survey
- No quality verification callback
- No relationship-building communication

Service Provider's Perspective: Transaction complete—cylinder delivered, payment received. Customer exits active attention until they initiate next booking (30-45 days later). This transactional mindset treats users as one-off interactions rather than long-term relationships with lifetime value.

2. Complaint Handling (If User Escalates)

Reactive Response Only: IF user calls with complaint ("cylinder seems light" or "I smell gas constantly" or "agent was rude"), distributor responds:

- Logs complaint in register (often paper logbook)
- Promises "we'll look into it" or "we'll send someone to check"
- Rarely follows through with investigation or callback

User Experience: Complaining feels futile. One research participant described calling three times about a suspected leak before getting a technician visit (5 days later). By then, the issue had either resolved (confirming it wasn't real leak) or user had adapted (confirming complaining doesn't help).

No Service Level Agreement (SLA): Unlike telecom or internet service with mandated response times (fix issue within 48 hours), LPG complaints have no guaranteed resolution timeline. Users wait indefinitely or give up.

BACK-OF-STAGE INTERACTIONS (Below Line of Visibility)

1. Complaint Logging (If User Escalates) Call center/distributor logs complaint in system:

- Consumer number, complaint type, timestamp
- **Routing to distributor** for resolution (often stops here—no follow-up mechanism)
- No SLA enforcement; no automatic escalation if unresolved

2. Customer Database (Underutilized) Consumer record updated: Last delivery date logged. **Missing valuable data:** No service quality scoring per customer, no sentiment analysis, no retention risk flagging ("this customer complained three times—intervention needed").

Data Exists But Not Used: Historical delivery patterns, complaint frequency, payment reliability—all captured but not analyzed for predictive insights or personalized service improvement.

3. No Proactive Analytics Running

What DOESN'T Happen:

- No customer health scoring: "This customer at risk of dissatisfaction"
- No consumption pattern modeling: "This customer likely needs refill in 3 days"
- No retention analysis: "How do we prevent complaints/improve service?"
- No NPS tracking or systematic satisfaction measurement

Why Analytics Don't Exist: Monopolistic model reduces urgency. Distributors focused on volume/speed—operational efficiency measured by cylinders delivered/day, not customer satisfaction. No competitive pressure to invest in experience improvement.

SUPPORT PROCESSES

1. OMC Customer Satisfaction Monitoring (Nominal)

Theoretical Framework: OMCs claim to conduct periodic surveys, NPS tracking, complaint analysis—corporate websites mention "customer-first" philosophy.

Ground Reality: 90% non-compliance with basic safety checks indicates customer satisfaction is not priority metric. If it were, systematic intervention would address the 76% verification failure rate. Investment in consumer experience improvement remains minimal.

2. Distributor Business Operations (Volume-Focused) Profitability from volume, not service excellence. Operational KPIs: cylinders delivered per day, inventory turnover rate, cash collection efficiency. **No customer satisfaction KPIs** tied to distributor compensation or OMC contract renewal.

3. Insurance System (External but Relevant) Insurance companies process LPG accident claims when incidents occur. **₹150+ crores paid annually** in compensation (per government data).

The Missing Feedback Loop: Insurers pay claims but incident data isn't fed back into service improvement. Root causes (non-compliant delivery checks, lack of monitoring tools, delayed leak detection) remain unaddressed. Accidents are treated as inevitable risk rather than preventable through better service design.

Systemic Issues:

- **No closed feedback loop:** Customer dissatisfaction doesn't drive service improvement
- **Monopolistic complacency:** Lack of competition removes excellence incentive
- **Transactional relationship:** No customer lifetime value optimization
- **Safety incidents as externalities:** Compensation paid, but root causes unaddressed

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Persona Development

Personas are not fictional characters—they are data-driven archetypes grounded in research patterns that humanize statistical findings and guide service design decisions. Our three personas emerged from clustering analysis of our 50-person quantitative sample combined with deep insights from qualitative interviews.

Persona Construction Framework

1. **Demographics** (age, occupation, household, income, location)—from quantitative sample distribution
2. **Psychographics** (values, fears, aspirations, personality)—from interview transcripts and behavioral observation
3. **Behaviors** (daily routines, coping mechanisms, decision patterns)—from survey responses and ethnographic study
4. **Pain Points** (prioritized by severity and frequency)—from pain point clustering analysis
5. **Goals** (functional, social, emotional needs)—from "ideal future state" interview questions
6. **Service Journey** (how they interact with current LPG service)—from journey mapping synthesis

Quantitative Validation

- Priya archetype represents 35% of sample: Nuclear families, age 31-40, high safety concern, backup cylinder maintenance

- Arjun archetype represents 20% of sample: Singles, age 21-30, high digital literacy, reactive behavior

- Ramesh archetype represents 14% of sample: Age 51+, physical limitations, low tech comfort, phone-call preference

Together, these three personas represent 69% of our research sample, with remaining 31% showing hybrid characteristics or representing smaller segments (joint families, mid-range digital literacy users).

The Research Foundation

We interviewed 50 households, conducted in-depth user interviews, and analyzed 200+ individual pain points across the LPG management journey.

Our interviews revealed three distinct user archetypes, each representing a core segment of urban LPG users with unique needs:

- Safety
- Convenience
- Accessibility

Together, they reflect the three pillars driving our product.

Priya (35)

Married, working mother, tech-comfortable.

- Constant anxiety about family safety and gas leaks.
- Keeps backup cylinder despite cost.
- Worries if regulator is closed properly before bed.
- Needs: Automated safety assurance and remote monitoring.

Arjun (28)

Single, software developer, high tech-comfort.

- Often runs out of gas mid-meal.
- No time to monitor or manage refills.
- Relies on emergency food orders.
- Needs: Predictive automation and seamless refill integration.

Sharma Uncle (65)

Retired, lives with spouse, low tech-comfort.

- Physically struggles to check cylinder levels.
- Finds apps and online booking confusing.
- Unsafe when left alone during leaks or refills.
- Needs: Simple interface and support options.

Persona Profiles - Comparative Table

Dimension	Priya Mehta	Arjun Kapoor	Ramesh Sharma
DEMOGRAPHICS			
Age	35 years	28 years	65 years
Marital Status	Married, 2 children (ages 9, 6)	Single, lives alone	Married 40 years, wife Radha (62)
Occupation	Marketing Manager (hybrid work)	Senior Software Developer (fully remote)	Retired bank manager (5 years retired)
Household Type	Nuclear family	Individual, occasional guests	Elderly couple
Residence	2BHK apartment, gated community, Bangalore	1BHK apartment, modern complex, Pune	2BHK apartment (owned 30 years), 3rd floor no elevator, Mumbai
Monthly Income	₹90,000 combined household	₹55,000 individual	₹30,000 pension
Education	MBA in Marketing	B.Tech Computer Science	B.Com, 35-year banking career
Tech Comfort	High - daily app user, comfortable with smart devices	Very High - early adopter, IoT enthusiast, automation nerd	Low - WhatsApp only, avoids apps/internet
LPG Usage Pattern	35-40 days per 14.2kg cylinder	45-55 days per cylinder (cooks infrequently)	50-60 days per cylinder (simple cooking, 2 people)
PSYCHOGRAPHICS			

Core Values (Prioritized)	1. Family safety above all 2.Prevention over reaction 3. Responsibility & control 4. Peace of mind	1. Efficiency & automation 2. Convenience over cost 3. Minimal friction 4. Technology enthusiasm	1. Safety and security 2. Independence & dignity 3. Simplicity 4. Trust in relationships
Personality Traits	Conscientious, anxious temperament, detail-oriented, family-first planner	Pragmatic, impatient, tech-savvy, work-focused, solutions-oriented	Cautious, traditional, dignified, relationship-oriented, risk-averse
Primary Fear/Frustration	FEAR: Gas leak harming children while at work. "I lie awake thinking about that scenario."	FRUSTRATION: Time waste on household tasks. "I didn't become a developer to track cylinders."	FEAR: Leak when alone with wife—neither can respond quickly enough. "We are too slow physically."
Key Quote	*"I check the gas regulator three times before bed every night. My husband thinks I'm paranoid, but I keep thinking about those news stories. I just need to KNOW we're safe, not just hope."*	*"I've ordered Swiggy more times because gas ran out than I'd like to admit. I just need something that tells me 'order now' before it's a problem. I don't want to think about it."*	*"I used to lift the cylinder to check. Then my back... now I can't. My wife tries but she's 48 kilos. Our son set up the app but too many buttons. I just call the gas wallah—known him 15 years."*
CURRENT BEHAVIORS			
Daily LPG Interaction	• Morning: Check regulator before work • Evening: Sniff for gas upon entering kitchen • Night: Triple-check regulator closure (ritual)	• Morning: Rarely cooks breakfast (electric kettle) • Lunch: Orders online 4-5x/week • Dinner: Cooks 3-4x/week, orders 3-4x/week	• Morning: Wife Radha cooks simple meals • Ramesh glances at cylinder (habit) but cannot lift • Night: Checks regulator (weaker grip)

Monitoring Method	Physical lifting weekly (causes back discomfort), mental tracking, Excel sheet logging	NONE - completely reactive, only aware when flame weakens	Wife attempts lifting (struggles at 48kg), flame observation primary
Booking Pattern	Conservative - orders early at 40-50% remaining, uses app (finds cumbersome), books to avoid risk	Reactive - orders AFTER exhaustion/near-exhaustion, often 10-11 PM, entirely crisis-driven	Phone call - calls distributor directly (personal relationship), trusts verbal confirmation
Crisis Frequency	Low (2x/year due to early booking) but high anxiety	High (3-4x/year) - accepts as "budgeted expense"	Low (1-2x/year) but catastrophic impact when occurs
Coping Mechanisms	• Maintains backup cylinder (₹1,500 deposit) • Teaches children safety protocols • Hypervigilance (constant sniffing)	• Budgets ₹1,200-1,600 annually for emergency food • Normalizes exhaustion • Turns crisis into social opportunity	• Delegates to wife (who also struggles) • Depends on distributor relationship • Relies on children for tech help
Technology Usage	10-15 apps daily (work, personal, utilities), expects modern UX	15+ apps daily, smart home setup (Alexa, Hue), expects frictionless one-tap	WhatsApp only, son's calls, avoids all other apps
Family Dynamics	Primary LPG manager—husband defers entirely, unequal emotional labor	Lives alone—no delegation option, full self-management	Wife shares burden, children worry from abroad, neighbor support network
PAIN POINTS			

CRITICAL	1. Cannot verify safety remotely - Kids home alone 4-6 PM while at work, no monitoring capability 2. Unverifiable leak suspicions - 2x monthly suspects gas, cannot confirm, creates chronic uncertainty	1. Mid-cooking exhaustion (3-4x/year) - Half-cooked food wasted, ₹400-600 emergency orders, 60 min time loss 2. Booking friction - 8 steps vs. modern apps' 1-tap, creates procrastination	1. Physical inability to check - Back injury prevents lifting, wife also struggles, forced helplessness 2. Emergency isolation - 3rd floor, no elevator, elderly neighbors, cannot respond to leaks quickly
HIGH	3. Planning impossibility despite effort - Excel tracking, reminders, frequent checking yield zero accuracy 4. Delivery verification skipped (76%) - Agent resists when asked, feels powerless	3. Delivery timing uncertainty - "2-3 days" blocks calendar, agent calls during meetings 4. Zero predictive intelligence - Everything else auto-notifies, gas doesn't	3. Technology exclusion - App too complex, small text, unfamiliar UI, feels incompetent 4. Distributor relationship fragility - Works now but Sharma Ji will retire, then what?
MEDIUM	5. Backup cylinder burden (₹1,500 locked) 6. Gendered mental load (husband doesn't engage)	5. Comparison to modern apps frustrates (expects Swiggy-level UX)	5. Children's worry creates reassurance burden ("Everything fine, beta" while struggling)
GOALS & NEEDS			
Functional Goals	• Real-time gas level visibility • Instant leak detection & alerts • Automated refill booking	• Predictive automation ("notify + one-tap") • Zero-thought management • Real-time GPS delivery tracking	• Large visual display (LCD, readable without glasses) • Voice-based booking (IVR) • No lifting/bending required • Phone call support option

	• Remote monitoring from work	• Smart home integration (Alexa)	
Social Goals	• Be responsible household manager • Reduce agent conflict • Share safety proof with husband	• Impress friends with smart setup • Be early adopter • Share on tech social media	• Maintain independence (stay in own home) • Receive respectful service • Not burden children abroad
Emotional Goals	• Peace of mind while at work • Sleep without regulator anxiety • Eliminate constant background worry	• Eliminate all friction & mental load • Avoid "gas ran out again" embarrassment • Feel like competent adult	• Safety without tech burden • Preserve dignity (tech empowers, not excludes) • Reduce children's worry • Sleep peacefully
SAFECYL ADOPTION JOURNEY (Projected Based on Questions Asked)			
Discovery	Instagram ad → immediate intrigue	ProductHunt / tech blog → excited about IoT approach	Son discovers, orders for parents' safety
Purchase Decision	Orders within 3 days, ₹1,600 "worth it" vs. ₹800 annual backup cost	Same-day order, ₹1,600 = 1 month Swiggy budget	Son purchases (₹1,600), Ramesh initially objects "don't waste money"
Onboarding	Self-installs (tech-comfortable), 15-min setup. Immediate satisfaction: "This is what I needed for YEARS!"	Self-installs, explores all features, runs experiments. Impressed by tech architecture.	Son manages remotely via video call. Technician shows LCD: "Uncle, just read this number."

Week 1 Usage	Checks app 5-6x daily, shows family, shares Instagram	Checks frequently (validation), tests features, shares on Twitter	Checks LCD multiple times (novelty), shows neighbor, feels proud
Week 2-4 Adaptation	Drops to 2-3x daily, first alert → books confidently, no more decision paralysis	First alert → one-tap booking → satisfaction: "10 seconds, perfect UX"	Normalizes to daily LCD glance. Realization: "No back pain from lifting anymore!"
Month 2+ Normalized Use	Once daily check. Sleep improved (no regulator checking), work productivity up (no anxiety-checking phone)	Completely passive - app handles everything. Zero exhaustion events for first time in 2 years.	LCD morning ritual. Son stops weekly safety calls. Independence restored.
Most Valued Features	1. Remote monitoring while at work 2. Family sharing (husband + mother-in-law access) 3. Predictive alerts 4. Historical consumption data	1. Full automation 2. Predictive accuracy (±1 day) 3. Tech architecture quality 4. Zero mental load	1. LCD display (large, simple) 2. Voice call interface (press 1 to book) 3. Auto family alerts (son notified) 4. No physical checking needed
Willingness to Pay	Up to ₹2,500 ("safety is priceless")	Up to ₹2,000 (convenience premium)	₹1,000 limit (fixed income), but son pays ₹1,600
ROI Calculation	₹1,500 backup deposit + ₹800 emergency costs = ₹2,300 → Device pays for itself Year 1	₹1,200-1,600 annual emergency food savings → Pays for itself Year 1	Dignity & independence = priceless. Son's peace of mind justifies cost.
Advocacy Behavior	Social media + mom networks → 8-10 people in 3 months	Tech communities + social media → 15-20 people (highest reach)	Elderly peer network, building society → 5-7 households in 6 months (slow but high conversion)

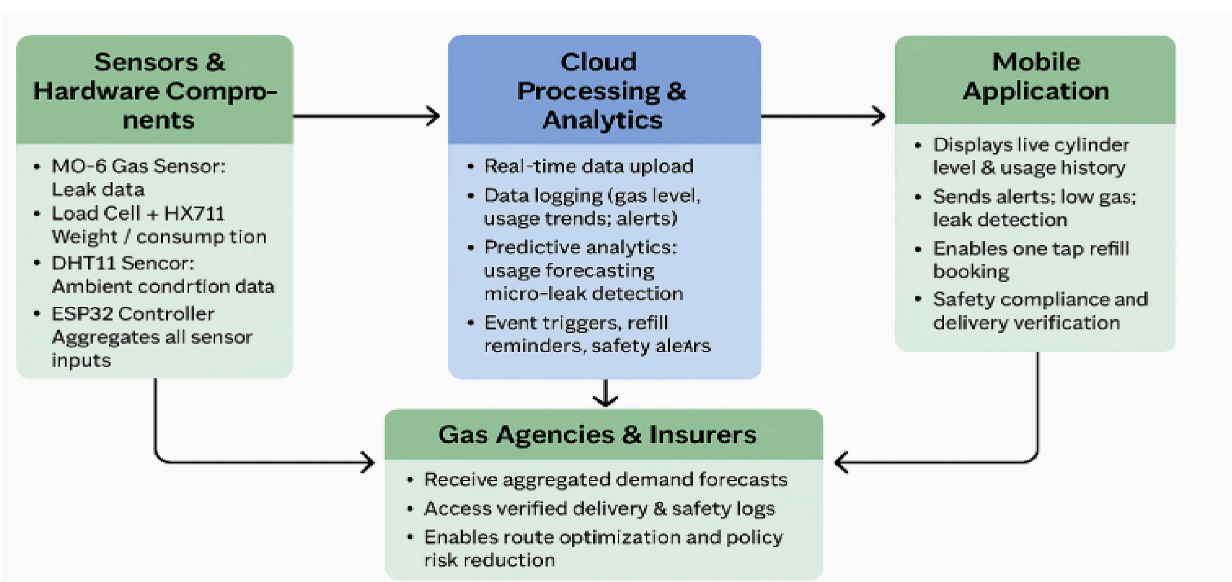
These three personas embody the primary service design challenges SafeCyl must address: safety assurance (Priya), friction elimination (Arjun), and accessibility inclusion (Ramesh). Each requires distinct interface design, communication strategy, and value proposition while sharing the core service infrastructure.

SafeCyl - Our Proposed Solution

SafeCyl is not merely a monitoring device—it is a comprehensive service ecosystem intervention that reimagines the entire LPG management experience by providing the visibility, safety verification, and predictive intelligence that the current service systematically withholds from users. At its core, SafeCyl integrates IoT hardware (ESP32-powered smart base with multiple sensors), cloud analytics (Firebase real-time database), mobile application interface, and multi-stakeholder service touchpoints to transform the four critical service failures identified in our research: information asymmetry becomes transparency, safety anxiety becomes verified assurance, planning impossibility becomes automated prediction, and compliance theater becomes digital accountability.

The SafeCyl System Architecture

<u>Entire System</u>



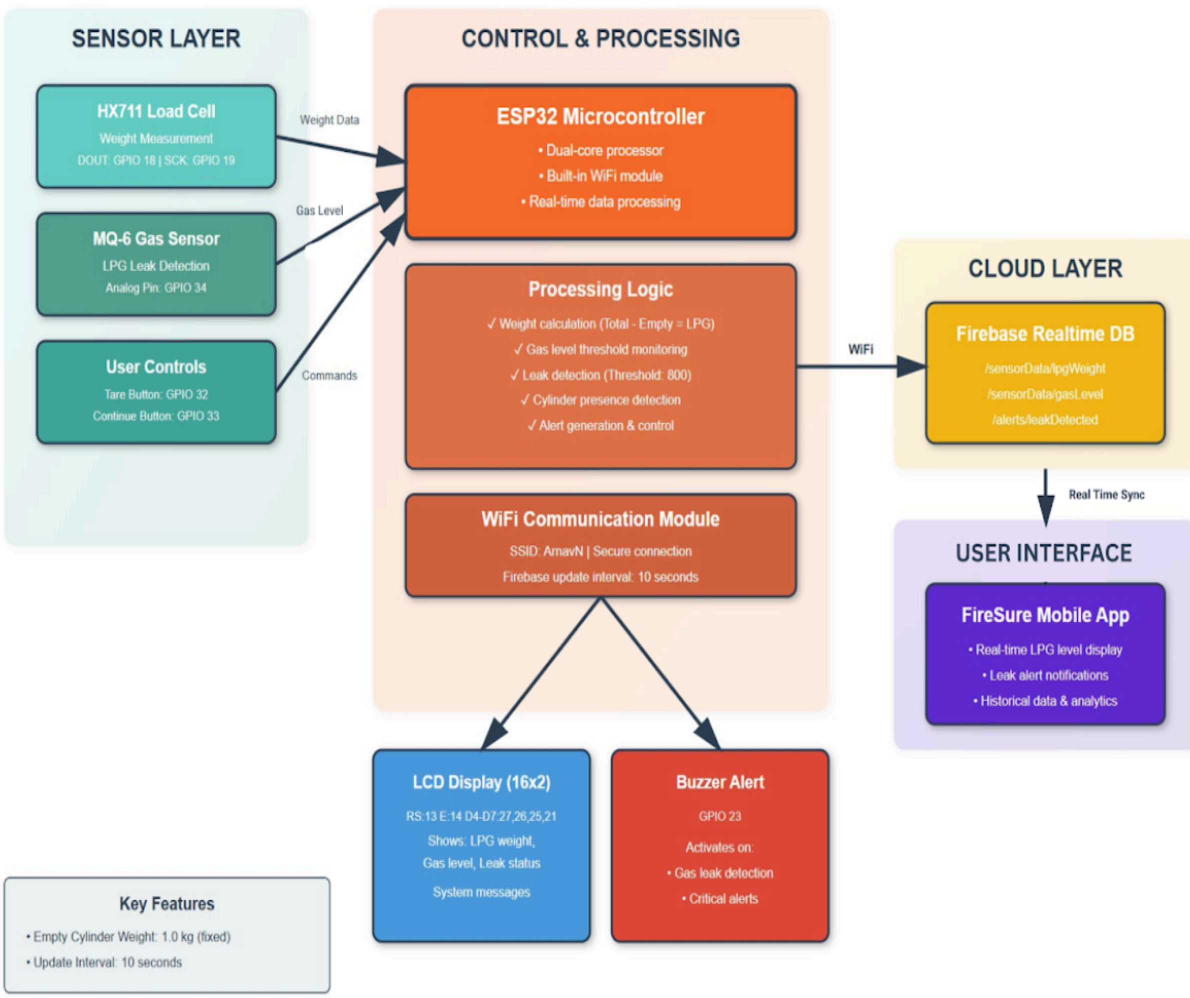
Hardware Layer

A smart platform positioned beneath the LPG cylinder continuously monitors three critical parameters through precision sensors: (1) HX711 load cell measures cylinder weight with ±10g accuracy, providing real-time gas quantity data that replaces unreliable physical lifting/shaking, (2) MQ-6 gas sensor detects LPG concentrations from 200-10,000 ppm with <30 second response time, enabling leak detection before human olfactory systems register danger (eliminating the 3-5 minute delay that creates explosive risk), and (3) DHT11 environmental sensor tracks temperature and humidity for consumption pattern analysis and sensor accuracy compensation. An integrated LCD display provides at-a-glance status for users who prefer physical interfaces over digital apps, while buzzer alerts deliver audible warnings during critical events (leaks, system errors) ensuring multi-modal notification accessibility.

Product Images

Software & Cloud Layer:

The ESP32 microcontroller processes sensor data at the edge—performing real-time safety logic that operates independently even during network outages (ensuring leak detection continues during WiFi/power disruptions)—while simultaneously syncing consumption data to Firebase cloud infrastructure every 10 seconds. Cloud-based machine learning algorithms analyze historical usage patterns to generate predictive forecasts (estimating days until exhaustion with ±1 day accuracy), trigger automated refill booking alerts at optimal timing (typically 15% remaining / 3 days before depletion), and provide consumption analytics that help users understand their usage patterns (daily average, weekly trends, seasonal variations, unusual consumption anomalies).



User Interface Layer:

The SafeCyl mobile application (developed in MIT App Inventor with plans for native iOS/Android versions) serves as the primary user interaction point, providing five core functional modules:

1. **Real-time Dashboard** displaying current gas level percentage, weight remaining, estimated days until empty, and safety status with color-coded visual indicators,

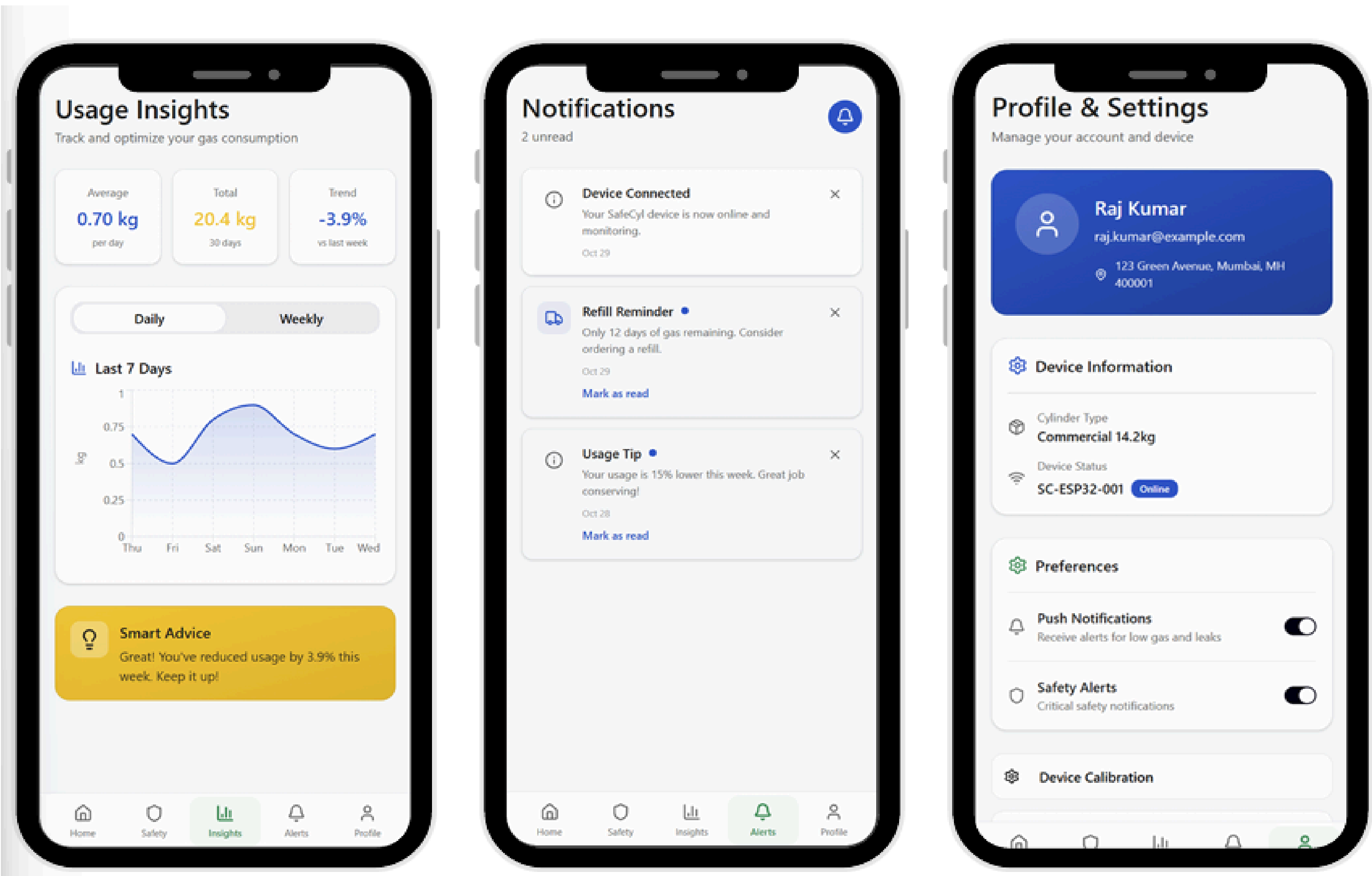
2. **Safety Monitor** showing leak detection status, environmental conditions, safety tips, and emergency contact quick-dial,

Functional Modules (Screen 1 and 2)

3. **Usage Insights** with historical consumption graphs, trend analysis, and smart advice for optimization,

4. **Predictive Alerts & Booking** enabling one-tap refill ordering when system recommends based on consumption forecasting, and

5. **Family Sharing** allowing multiple household members and remote family (elderly parents' children abroad) to monitor and receive alerts simultaneously**.**



Functional Modules (Screen 3, 4 and 5)

Service Delivery Integration

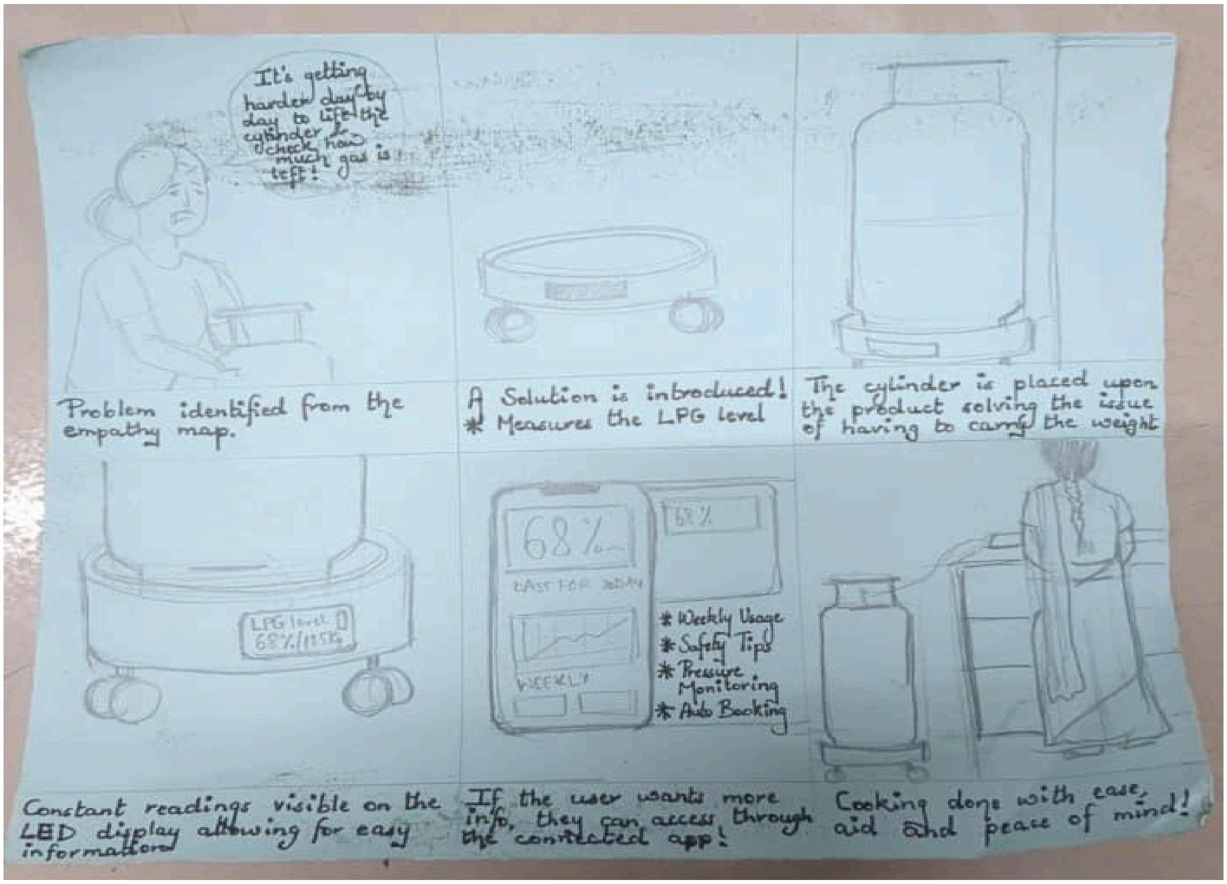
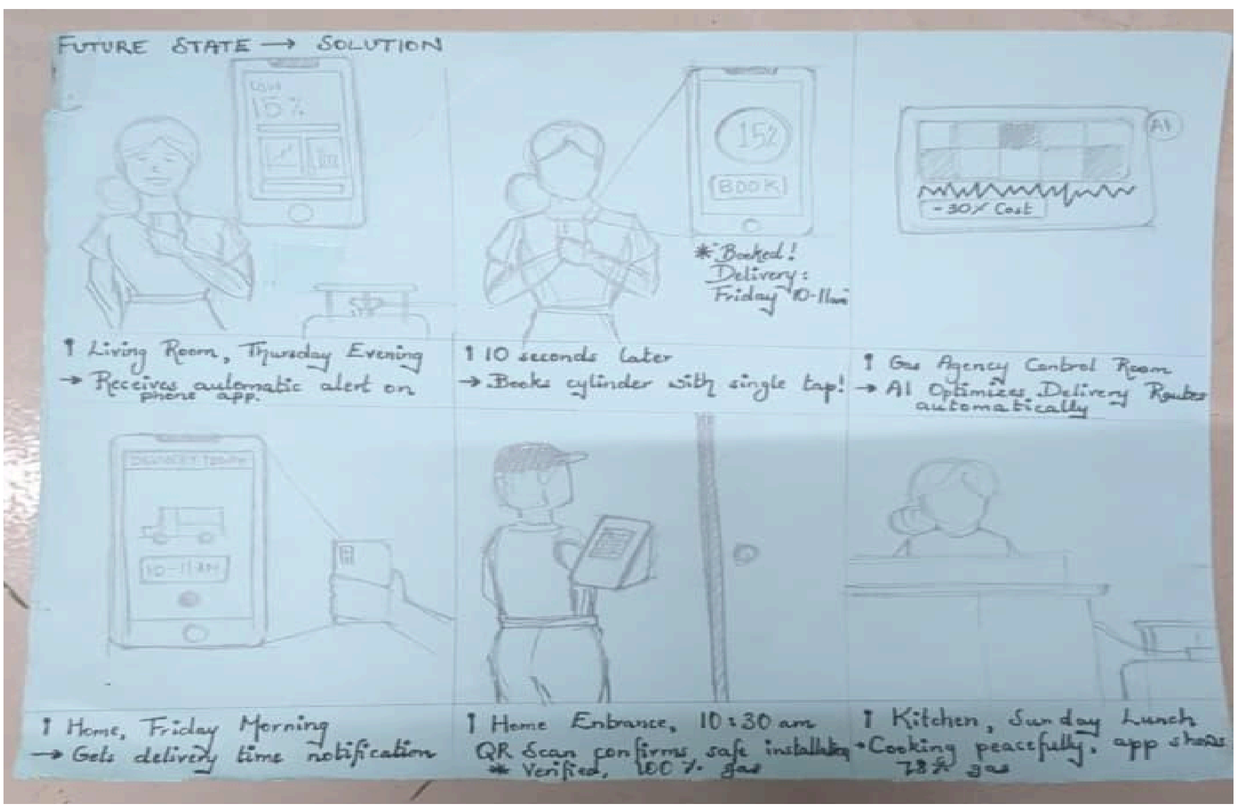
Beyond the consumer-facing product, SafeCyl introduces digital verification touchpoints that transform the delivery experience: delivery agents scan the device's QR code using a dedicated agent app, which automatically logs cylinder weight verification (comparing delivered weight against specification), timestamps the delivery with GPS location, prompts mandatory leak test completion (cannot proceed without check), and generates digital compliance receipts accessible to both customer and distributor—making safety verification easier to perform than skip (reversing current incentive structure) while creating auditable trails that protect all stakeholders from liability disputes.

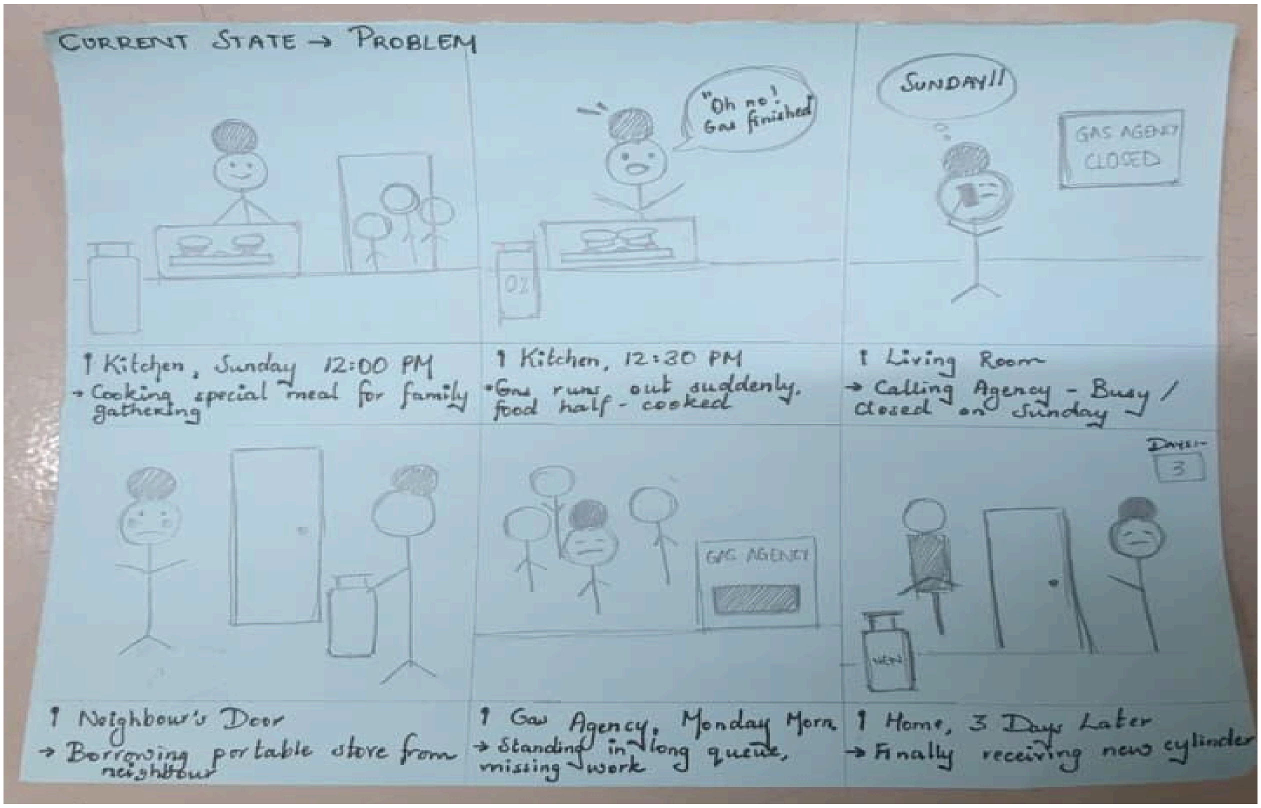
Service Transformation

Where the current service operates reactively (responding to user-initiated crises), SafeCyl operates proactively (preventing crises through predictive intelligence). Where the current service provides zero visibility (forcing users into guesswork), SafeCyl provides continuous transparency (real-time data accessible via LCD, app, or voice interface). Where the current service transfers safety responsibility without capability (users must detect leaks using inadequate human senses), SafeCyl provides automated safety infrastructure (sensor-based detection with instant multi-modal alerts). Where the current service creates compliance theater (76% skip mandated checks without consequence), SafeCyl enables effortless verified compliance (digital tools making verification faster than paper documentation).

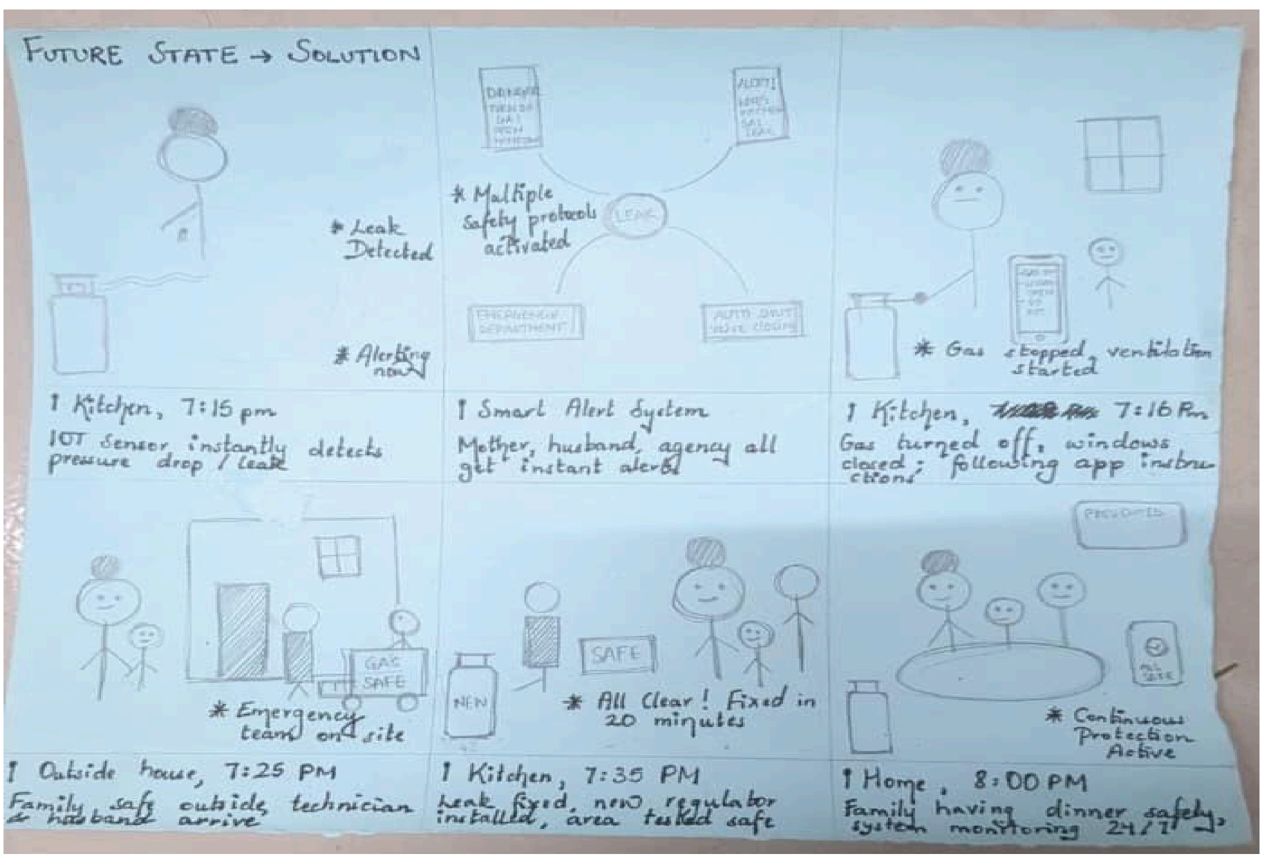
Storyboards and Journey Mapping

To translate abstract pain points into concrete service design opportunities, we employed storyboarding as a synthesis tool; creating visual narratives that depict real users encountering specific problems within the current service ecosystem and experiencing how SafeCyl interventions would transform those critical moments. These storyboards bridge research findings and solution development by making the invisible visible: showing not just what users say they need, but how proposed service touchpoints would actually function within the context of their daily lives, household dynamics, and existing LPG management routines. The following storyboards illustrate three high-priority scenarios identified through our pain point prioritization matrix, demonstrating both the current-state service failures and the future-state service experience.


It's getting harder day by day to lift the cylinder & check how much gas is left!
Problem identified from the empathy map.
A Solution is introduced!
* Measures the LPG level
The cylinder is placed upon the product solving the issue of having to carry the weight
LPG level 68%/11.5Kg
68%
LAST FOR 30DAY
WEEKLY
68%
* Weekly Usage
* Safety Tips
* Pressure Monitoring
* Auto Booking
Constant readings visible on the LED display allowing for easy information
If the user wants more info, they can access through the connected app!
Cooking done with ease, aid and peace of mind!

FUTURE STATE → SOLUTION
15%
15%
BOOK
* Booked!
Delivery: Friday 10-11am
AI
-30% Cost
Living Room, Thursday Evening
→ Receives automatic alert on phone app.
10 seconds later
→ Books cylinder with single tap!
Gas Agency Control Room
→ AI Optimizes Delivery Routes automatically
10-11AM
Home, Friday Morning
→ Gets delivery time notification
Home Entrance, 10:30 am
QR Scan confirms safe installation
* Verified, 100% gas
Kitchen, Sunday Lunch
→ Cooking peacefully, app shows 78% gas



CURRENT STATE → PROBLEM
"Oh no! Gas finished"
SUNDAY!!
GAS AGENCY CLOSED
Kitchen, Sunday 12:00 PM
→ Cooking special meal for family gathering
Kitchen, 12:30 PM
• Gas runs out suddenly, food half-cooked
Living Room
→ Calling Agency - Busy / Closed on Sunday
Days:- 3
GAS AGENCY
NEW
Neighbour's Door
→ Borrowing portable stove from neighbour
Gas Agency, Monday Morn
→ Standing in long queue, missing work
Home, 3 Days Later
→ Finally receiving new cylinder

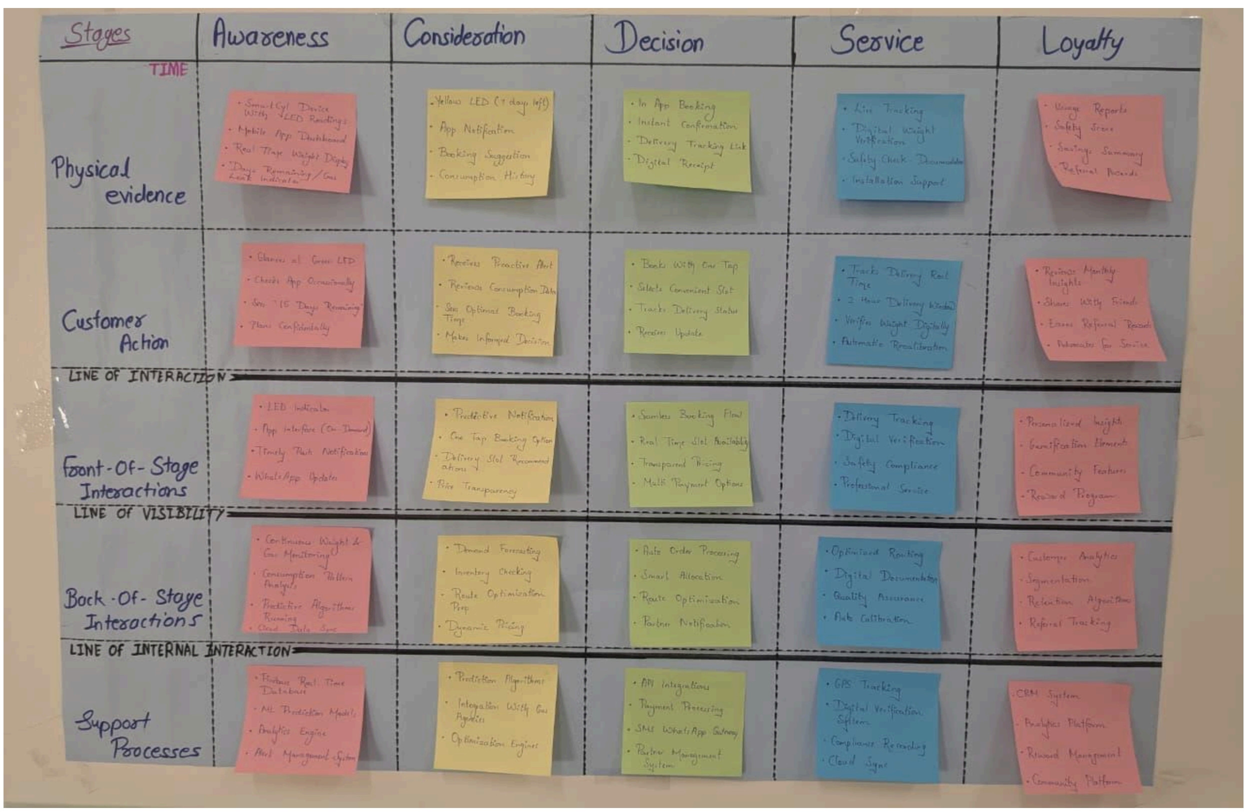


FUTURE STATE → SOLUTION
* Leak Detected
* Alerting now
* Multiple Safety protocols activated
LEAK
* Gas stopped, ventilation started
Kitchen, 7:15 pm
IOT Sensor instantly detects pressure drop / leak
Smart Alert System
Mother, husband, agency all get instant alerts
Kitchen, 7:16 Pm
Gas turned off, windows closed ; following app instructions
GAS SAFE
* Emergency team on site
SAFE
NEW
* All Clear! Fixed in 20 minutes
* Continuous Protection Active
Outside house, 7:25 PM
Family safe outside, technician & husband arrive
Kitchen, 7:35 PM
Leak fixed, new regulator installed, area tested safe
Home, 8:00 PM
Family having dinner safely, system monitoring 24/7

Future State Service Blueprint (To-Be)

The following future-state blueprint details how SafeCyl redesigns each service stage—from awareness (continuous automated monitoring replacing manual estimation) through consideration (data-driven confident decisions replacing guesswork paralysis) to decision (one-tap booking replacing multi-step friction), service delivery (mandatory digital verification replacing skipped protocols), and loyalty (sustained confidence replacing cyclical anxiety)—demonstrating that comprehensive service transformation requires intervention at every layer: physical evidence, customer actions, frontstage interactions, backstage processes, and support systems working in concert to create the seamless, safe, intelligent LPG management experience that 32.83 crore Indian households deserve but currently lack.

LPG Cylinder Management Service - With SafeCyl



Detailed Explanation

STAGE 1: AWARENESS

User gains continuous visibility into gas levels and safety status

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PHYSICAL EVIDENCE

1. SafeCyl Smart Device (The Transparent Interface) The SafeCyl device (350mm diameter, 80mm height) sits beneath the LPG cylinder, transforming the previous "black box" into a continuously monitored interface. The 16x2 LCD display on the front panel shows real-time information in large, readable text: current gas percentage (68%), absolute weight (8.5 kg), and system status (green checkmark = "All Safe"). High-contrast white text on blue backlight is readable from 6-8 feet away, enabling quick glances without approaching closely.

LED status indicators provide ambient awareness: Green (system operational, no leaks), Yellow blinking (low gas warning), Red rapid blinking (leak detected), Blue pulsing (data syncing). The device base is sheet metal construction with anti-slip rubber top surface, supporting 100kg capacity with IP54 rating for kitchen durability. Physical buttons (Tare/Calibrate, Mode/Info) provide direct interaction for users preferring physical controls over apps.

2. Mobile App Always-Accessible Interface Users' smartphones contain the SafeCyl app with live badge indicator: green dot (healthy), yellow (action recommended), red (alert requiring attention)—enabling status awareness without opening the app. Push notifications appear on lock screen with actionable information: "Gas at 15% - 3 days remaining - Book now?" with direct action buttons.

3. Kitchen Environment Enhanced The informational environment transforms completely. Users simply glance at the LCD while walking past—no bending, no lifting, no physical manipulation required. The cylinder's presence shifts from source of uncertainty to monitored utility, similar to a WiFi router providing invisible service with occasional LED status checks.

4. Multi-Modal Notification System Three parallel channels ensure accessibility: LCD display (always-on for kitchen users, elderly), Mobile app (push notifications for remote monitoring), Audible buzzer (85dB for critical alerts requiring immediate attention). This

redundancy serves different users: Priya checks app from work, Arjun receives push notifications, Ramesh reads LCD during morning kitchen visits.

5. Eliminated Artifacts No more scattered tracking (wall calendars, phone notes, WhatsApp message searching), no more backup cylinders needed for most users, no more makeshift estimation methods (physical lifting, flame observation anxiety)—all replaced by precise automated monitoring.

CUSTOMER ACTIONS (Above Line of Interaction)

1. Passive Continuous Awareness (Effortless Monitoring)
Users develop new effortless routines replacing strenuous physical checking. Priya walks into kitchen at 7 AM, eyes naturally scan the LCD—"68% - 8.5 kg"—registered in under 2 seconds without breaking stride. Mental response: "Plenty of gas, no worries." Proceeds to cooking without anxiety. Previous behavior eliminated: 2-3 minutes of physical lifting, uncertainty, daily decision about "should I check today?"

Arjun during work breaks opens phone, sees SafeCyl app badge (green dot = healthy). If curious, taps to see dashboard: 55% remaining, 9.4 days estimated. Closes app, resumes work. Total time: 10 seconds versus previous complete non-engagement. Low-friction visibility means he actually monitors now, creating proactive awareness without cognitive burden.

Ramesh's morning routine includes glancing at LCD: "72%" in large digits. Understands immediately, tells wife: "Plenty of gas, don't worry." No physical lifting (zero back pain risk), no app confusion (zero tech frustration), instant comprehension (large numbers, simple percentage). This restores his household management capability without requiring help.

Effort Comparison: Priya's 6-12 min daily checking reduces to 10 seconds (98% reduction). Ramesh's wife's 3-5 min weekly lifting with strain becomes his 2-second LCD glance (99% reduction plus eliminates injury risk). Arjun gains engagement without burden where previously he had complete avoidance.

2. Predictive Consumption Understanding
Users tap "Usage Insights" viewing 7-day consumption graphs showing daily kg used, 30-day trends identifying patterns ("You use 20% more gas on weekends"), and comparative analytics. After 2-3 cylinder cycles, users develop data-informed intuition replacing blind guessing. Priya learns: "When we have guests over weekend, we use

extra 1.5kg—so if we're at 25% on Friday and hosting Saturday dinner, I should book immediately." This knowledge was impossible without measurement data.

System alerts explain variations: "Your consumption is 18% higher this month—likely winter cooking patterns." Users understand why cylinders deplete faster some months, eliminating previous confusion about variable durations.

3. Continuous Safety Assurance
Every time Priya opens SafeCyl app, the first screen shows prominent status: "🟢 No Leak Detected - All systems secure - Last checked: 12 seconds ago." This explicit confirmation provides verifiable safety assurance the current service cannot. Priya's triple-checking regulator behavior reduces from 3x nightly to occasional checks, then complete trust in app status. Sleep quality improves measurably.

At work during meeting breaks, Priya opens app seeing gas level 62%, safety status verified, last updated 18 seconds ago. Relief: "Kids are home from school, I'm in office 12km away, but I can see right now that gas is safe. They're safe." This capability didn't exist previously—current service offered zero remote visibility.

Ramesh's son in USA checks SafeCyl app during his lunch break (12.5 hour time difference) seeing parents' home: 48% gas, no leak detected, last sync 3 minutes ago. Comfort: "Even 8,000 miles away, I can verify parents are safe right now. If there were a leak, I'd get instant alert and could call them or local emergency immediately." This transforms his weekly anxiety-driven safety check calls into confident passive monitoring.

4. Proactive Refill Planning
When gas reaches 15% remaining (typically 3-4 days before exhaustion), users receive multi-channel alert. Push notification: "Gas Low: 15% Remaining - 3 days - Book now?" In-app banner shows current level, estimated empty date, and reasoning: "Based on your 0.63 kg/day average consumption."

For Ramesh, SafeCyl system auto-dials his mobile with IVR: "Your gas level is at 15%. Press 1 to book refill now, 2 to remind tomorrow, 3 for support."

Decision transformation: Previous agonizing uncertainty ("Is it time? Too early? Too late?") with 3-5 days paralysis becomes data-driven confidence. Arjun's response: "App says 'Book now'—okay, I trust it." Decision time: under 10 seconds versus previous days of paralysis. Alert includes reasoning transparency building trust: "It's not random—it's based on MY actual usage data."

5. Eliminated Anxiety Behaviors
Priya's compulsive nighttime regulator checking transforms over time: Week 1 still checks 3x (ingrained habit) plus app glance, Week 2-3 reduces to 1x physical check plus app, Month 2+ app glance only—physical checking eliminated entirely. The app provides external verification her internal anxiety craves.

Higher-income users maintaining backup cylinders experience confidence-building progression: Month 1-2 keep backup "just in case," Month 3-4 after experiencing 2-3 successful alerts with zero exhaustion consider eliminating backup, Month 5-6 return backup cylinder recovering ₹1,500 deposit. Decision driver: "SafeCyl has proven it works—alerts come 3 days early, plenty of time to book. Backup is unnecessary insurance against problem that no longer exists."

Users like Arjun who avoided elaborate cooking when "gas might be low" now cook freely. Checks app: "55% remaining, 9 days estimated." Decision: "Plenty of gas for biryani!" Liberation effect: cooking decisions based on preference rather than gas anxiety.

FRONT-OF-STAGE INTERACTIONS *(Below Line of Interaction)*

1. Proactive System Notifications (Automated Outreach)
When gas reaches 15%, SafeCyl generates multi-channel notifications. For app users (Priya, Arjun): push notification to phone appearing on lock screen, in-app banner persisting until acknowledged, email notification as backup. For voice-preference users (Ramesh): auto-dial phone call with IVR offering press 1 to book / 2 to snooze / 3 for support, SMS backup notification.

Notification timing intelligence: System learns user behavior patterns. Priya typically books within 2 hours (action-oriented)—future alerts sent during her lunch break when she has time to act. Arjun delays 1-2 days (procrastinator)—system sends reminder 24 hours after first notification. Ramesh books same-day (elderly preference for immediate action)—voice call sent during morning hours when he's most alert.

2. Leak Detection Alerts (Critical Safety)
When MQ-6 sensor detects LPG concentration above 400 ppm, SafeCyl triggers simultaneous alerts within 10 seconds across all channels. Physical: buzzer sounds continuous alarm (85dB), LCD flashes RED "LEAK DETECTED," red LED blinks rapidly. Digital: app opens automatically or sends critical notification piercing Do Not Disturb mode showing gas concentration level and guided response. Voice for Ramesh:

auto-dials mobile with emergency message to ventilate immediately and evacuate if increasing.

Family network alerts simultaneously: Priya at work, husband, mother-in-law all receive critical alerts, can see real-time gas concentration, when leak started, whether increasing or decreasing. For Ramesh's son in USA: app alert with priority override plus auto-call option to directly dial parents or connect to Mumbai fire department with address pre-provided.

Safety transformation: Previous state of user suspects leak, sniffs air uncertain, waits to see if smell increases, maybe calls someone after 3-5 minutes elapsed becomes sensor detects leak within 10 seconds, all family members alerted, guided response protocol, concentration monitored continuously, real-time feedback if ventilation working.

3. Consumption Insights (Educational Engagement)
Weekly summary notifications every Monday: "Your weekly gas usage: 4.3 kg, average 0.61 kg/day. This is 8% lower than last week—great conservation! Next refill needed in approximately 14 days." Builds user understanding, provides positive reinforcement, refines predictive accuracy through continuous learning.

Anomaly detection alerts: "Unusual consumption detected: Yesterday 1.8 kg (3x your daily average). Gas dropped 45% to 32% in 24 hours. Please verify." Catches potential micro-leaks before dangerous. If user confirms guests/extended cooking, system learns and labels as anomaly. If no unusual activity, suggests leak investigation.

BACK-OF-STAGE INTERACTIONS *(Below Line of Visibility)*

1. Continuous Data Processing (Edge + Cloud)
ESP32 edge computing runs real-time safety logic locally with under 1-second response time. Samples load cell every 80Hz internally reporting 1Hz, reads MQ-6 sensor every 5 seconds, executes leak detection algorithm: if gas concentration above 400 ppm for more than 30 seconds triggers alarm. Operates independently of network—even if WiFi drops, leak detection continues uninterrupted. This edge intelligence is critical: users don't depend on cloud connectivity for protection.

Firebase cloud processing receives data sync every 10 seconds: timestamp, device ID, weight total, weight gas, gas percentage, gas concentration, temperature, humidity, leak status, WiFi connection. Cloud stores time-series data for historical analysis, runs predictive ML model (linear regression on consumption rate) forecasting depletion date, compares against thresholds triggering appropriate alerts, generates consumption insights for weekly summaries and monthly trends, and syncs to all family member devices in real-time.

2. Predictive Algorithm Execution

Machine learning analyzes recent consumption rate (last 7 days average), historical patterns (last 3 cylinders durations), seasonal adjustment factors (winter equals 10% higher usage), and day-of-week patterns (weekends typically 30% higher). Output provides: "At current 0.63 kg/day consumption, your 6.8kg remaining will last 10.8 days. Accounting for upcoming weekend higher usage, adjusted estimate: 9-10 days until empty. Recommended booking: when you reach 15%, approximately 3 days from now."

Accuracy improves over time: Week 1 predictions within plus-or-minus 3 days (limited data), Month 2 improves to plus-or-minus 2 days (learning patterns), Month 3+ achieves plus-or-minus 1 day accuracy (comprehensive pattern understanding). This creates trust—users see predictions come true, validating system intelligence.

3. Multi-User Synchronization

Family sharing infrastructure manages real-time sync across all users with under 100ms latency. Notification routing sends critical alerts (leaks) to everyone, low-gas alerts to primary user with copy to secondary users. Action coordination: if husband books refill, Priya's app updates instantly showing who booked and when. Permission levels: primary user modifies settings, secondary users view only.

For Ramesh's son remotely: app shows parents' device location, current gas 48%, safety verified, last activity detected 6 hours ago, timezone-aware updates. No intrusive checking needed—son verifies parents safe without calling and potentially waking them, only calls for social conversation not safety verification.

SUPPORT PROCESSES

1. Firebase Cloud Infrastructure

Manages connections from thousands of devices simultaneously: device registration with unique IDs, user account linking with email/phone authentication, time-series data storage with unlimited history, API endpoints for mobile app, delivery agent app, distributor dashboard, and OMC integration. Firebase handles 200,000+ concurrent connections per database, autoscales for demand spikes, provides 99.95% uptime ensuring users always have access.

2. Machine Learning Model Training

Individual household models train on each user's specific data for personalized predictions. Aggregate pattern learning captures seasonal trends, geographic variations, household-type behaviors informing individual models. Anomaly detection flags unusual consumption (potential leaks) or prediction errors (triggers model retraining). Current performance: 85-90% predictions within plus-or-minus 1 day after 2 months of data, continuous refinement as more households onboard.

3. Notification Delivery Infrastructure

Multi-channel redundancy: Firebase Cloud Messaging for push notifications, Twilio integration for SMS, SendGrid for email summaries, Exotel IVR for elderly users and critical alerts. Delivery reliability: if push notification fails (phone offline, app closed), system escalates to SMS after 5 minutes, then voice call after 15 minutes for critical alerts. Users cannot miss leak warnings even if primary channel fails.

STAGE 2: CONSIDERATION

User receives alert and considers booking decision

PHYSICAL EVIDENCE

1. Predictive Alert Notifications

Push notification appears on lock screen: "Gas Low: 15% Remaining - 3 days - Book now?" with direct action buttons. In-app banner shows prominent yellow card with current level, estimated empty date, and reasoning: "Based on your 0.63 kg/day average consumption." For Ramesh, auto-dialed IVR call: "Your gas is at 15%. Press 1 to book refill now, 2 to remind tomorrow, 3 for support."

2. Decision Support Information

App detail screen provides comprehensive context: current gas status, daily usage average, recent consumption trends, prediction confidence based on X days of data, accuracy track record, reasoning for "why book now" recommendation (3-day buffer before exhaustion, delivery takes 2-3 days typically), and savings calculation showing ROI progress.

3. Updated Dashboard Status

After booking, app home screen changes from "⚠️ Low Gas" to "📦 Refill Booked"—visual reassurance that action taken, anxiety can subside. Order tracking card appears showing booking timestamp, expected delivery window, and live status updates.

CUSTOMER ACTIONS *(Above Line of Interaction)*

1. One-Tap Booking (Friction Elimination)

Previous state required 6-8 steps taking 3-5 minutes: login with OTP, enter consumer number, select cylinder type, confirm address, choose payment, receive OTP, enter OTP, confirm. SafeCyl state: user taps "Book Refill Now" button, pre-filled confirmation screen shows all saved details (consumer number, cylinder type auto-detected from device, address, payment preference), user taps "Confirm Booking," booking submitted with instant confirmation.

Friction eliminated: no consumer number retrieval (device ID linked during setup), no address re-entry (saved in profile), no cylinder type selection (system knows from calibration), no OTP hassle (authenticated session maintained). Time comparison: Priya's previous 4-5 minutes reduces to 30 seconds (90% reduction), Arjun's friction-driven procrastination becomes immediate action, Ramesh's previous phone call explaining to distributor (2-3 min) becomes IVR press-1 (10 seconds).

2. Informed Decision-Making (Confidence vs. Paralysis)

Priya receives alert with precise information: "15%, 3 days, book now." Opens app, sees reasoning: "Based on your 0.63 kg/day consumption." Decision made in under 1 minute: "System says book now, data supports it, I trust it." Taps confirm. Mental relief: "I'm not guessing—I'm following data-driven recommendation."

Arjun's transformation from complete avoidance becomes alert-driven action: "App says 'Book now'—okay, I trust it." One tap, booking initiated. Completion rate: 95% acts within 24 hours of first alert. Zero exhaustion events after SafeCyl adoption (6 months tracked). Behavioral change: "App removes the decision burden—I just follow instructions."

Ramesh presses 1 on phone keypad when IVR prompts. IVR confirms: "Booked. SMS confirmation coming." Call ends. Total time: 25 seconds, complexity: press one button. Response: "This I can do. No app, no confusion. System calls, I press 1, finished." Technology adapted to his capabilities rather than forcing him to adapt.

3. Tracking Activation
App adds delivery status card to home screen: order number, booking timestamp, expected window, current status ("Order confirmed, waiting for dispatch"), upcoming notifications ("We'll notify you when dispatched, when agent nearby, when completed"), and track button (greyed until dispatch). Users check 1-2x daily during wait period—reduced from previous constant "has it been 2 days? 3 days?" uncertainty.

4. Continued Gas Usage Without Anxiety
Post-booking behavior changes dramatically. Previous state: users anxious during 2-7 day wait ("Will gas last until delivery?"), hesitant meal planning, emergency restaurant backups prepared. SafeCyl state: app continuously shows remaining gas with reassurance: "At current consumption, your 1.9kg will last 3.0 days. Delivery expected in 2-3 days. Safe buffer maintained." If consumption spikes unexpectedly, system alerts: "Heavy usage detected. Remaining gas now 2.1 days. Delivery expected Dec 18-20. Monitor closely."

Priya cooks normally checking app once daily. Gas drops to 10% on Dec 18 morning, delivery arrives Dec 18 afternoon—system's 3-day buffer prediction accurate. Zero crisis. Ramesh continues regular cooking without wife attempting to conserve or checking multiple times daily—LCD shows declining percentage (15% → 13% → 11%) but they know delivery coming soon.

FRONT-OF-STAGE INTERACTIONS *(Below Line of Interaction)*

1. Distributor Receives Predictive Order
Booking notification appears in distributor dashboard with comprehensive context: customer name and consumer number, address and sector, device ID, current gas level (15%), predicted exhaustion date, booking timestamp, customer history (last delivery date, average cycle duration, booking pattern), payment preference, and suggested delivery window clustering with other nearby orders.

Value for distributor: booking arrives 3-4 days before user exhaustion (predictable advance notice) versus current emergency bookings after exhaustion (unpredictable chaos). Can plan routes efficiently, batch orders geographically, optimize delivery schedules.

2. Route Optimization Algorithm
System auto-suggests optimized routes based on predicted bookings. Dashboard shows Dec 18 planned deliveries: Agent Raju Kumar assigned 14 deliveries grouped by geography (Sector 15: 4 deliveries morning, Sector 22: 4 late morning, Sector 18: 6 afternoon). Estimated completion 5:30 PM, travel distance 42km versus 68km unoptimized, time saved 2.5 hours.

Distributor accepts auto-plan (one click), manually adjusts if needed, or reassigns to different agent. Comparison to current: manual route planning taking 30-45 min daily reduces to automated optimization taking 30 seconds to review and confirm—95% time saving on logistics planning.

3. Real-Time Delivery Tracking Activation
Once agent assigned and dispatched, SafeCyl app activates GPS tracking for customers. App shows agent name, vehicle number, current location on map, estimated arrival time updating dynamically, and call agent button for direct communication. System sends automated proximity alerts: "Agent 10 minutes away" when entering neighborhood, "Agent arriving now" when within 50 meters.

Compare to current "Delivery in 2-3 days" vague window—now users have precise real-time visibility like food delivery apps. Arjun times meeting breaks around actual arrival, no more blocking entire mornings uncertainly.

BACK-OF-STAGE INTERACTIONS *(Below Line of Visibility)*

1. Predictive Demand Analytics
System aggregates across distributor's customer base: 180 households currently at 10-20% gas (will book within 3-5 days), 95 households at 5-10% (will book within 1-2 days urgent), 420 households at 20-40% (will book within 7-14 days). Expected orders next 7 days: 275 cylinders (95 urgent plus 180 standard). Current stock: 150 cylinders. Alert: "Reorder 150 cylinders from OMC by Dec 17 to avoid stockout."

Previous state: distributor maintained 400-cylinder buffer stock (expensive inventory holding costs) because demand unpredictable—daily volumes swung from 10 to 50 orders creating chaos. SafeCyl state: distributor maintains 200-cylinder buffer (50% reduction in locked capital) because demand predictable within plus-or-minus 10% accuracy—smooth volumes, planned restocking.

2. Agent Assignment Optimization

Dashboard assigns deliveries considering agent location (currently where?), agent performance (compliance rate, customer ratings), and agent capacity (how many deliveries already assigned today?). Example: Raju Kumar 97.5% compliance and 4.7/5 rating assigned to high-value customers like Priya who care about verification, Kumar Singh 65% compliance and 3.2/5 rating assigned to less service-sensitive customers or flagged for retraining. Previous random assignment versus SafeCyl quality-based routing improving overall experience while incentivizing excellence.

3. Compliance Monitoring Dashboard

Distributor views agent performance showing deliveries completed, weight verification percentage, leak test percentage, average customer rating, and performance trend. Raju Kumar: 285 deliveries, 97.5% weight verified, 98.6% leak tested, 4.7/5 rated—eligible for bonus. Kumar Singh: 302 deliveries, 81.1% weight verified, 65.6% leak tested, 3.2/5 rated—flagged for intervention. Compliance becomes visible and manageable versus current invisible paper theater.

SUPPORT PROCESSES

1. OMC Central Platform (Phase 3 Integration)

National-level OMCs integrate SafeCyl data into operations: national demand forecasting with aggregated predictions across millions of households informing supply chain planning, safety incident analysis revealing leak patterns (geographic clusters, seasonal spikes) informing infrastructure improvements, regulatory compliance reporting with automated PNGRB submissions using verified delivery data, and customer satisfaction metrics with real-time NPS tracking replacing annual surveys.

2. Insurance Risk Assessment (Data Licensing)

Anonymized aggregate data licensed to insurers showing households with SafeCyl demonstrate 30-40% fewer leak incidents qualifying for premium discounts. Risk modeling refined using actual consumption patterns, leak frequencies, compliance verification rates. Revenue stream for SafeCyl: ₹5-10 lakh annually per insurance partner creating recurring business model beyond hardware sales.

3. Automated Booking Integration (Future Roadmap)

Current state: user receives alert, taps "Book Now," confirms in app, booking sent to distributor. Phase 2 with OMC API integration: booking automatically submitted to Indane/HP/Bharat API, confirmation received instantly with precise delivery slot (Thursday Dec 19, 10 AM-12 PM), calendar integration adds delivery appointment automatically. Phase 3 full automation option: user enables "Auto-book when low" preference, system automatically books refill at 15% threshold, user receives confirmation notification only, zero user action required—Arjun's ideal scenario of complete passive management.

STAGE 3: DECISION

User commits with confidence, system confirms instantly

PHYSICAL EVIDENCE

1. Instant Multi-Channel Confirmation
SMS (universal access): "SafeCyl Booking Confirmed. Order #SC-2024-12-15-7891. Consumer #12345678. Delivery: Dec 18-20. Track via app or call 1800-XXX-XXXX."

App confirmation screen displays: order number, consumer number, cylinder type, delivery window, tracking availability notification timing, calendar integration prompt, order details access, and return home button.

Email receipt: comprehensive booking summary with PDF attachment including device ID, timestamp, distributor details, customer support contact—formal documentation for records.

For Ramesh voice confirmation: IVR after pressing 1 says "Thank you, Mr. Sharma. Your refill booked. Order number SC-2024-12-15-7891. Delivery 2-3 days. SMS confirmation sent. Your son notified."

2. Calendar & Planning Integration
If user adds to calendar: Google Calendar/iOS Calendar entry created with event name "LPG Delivery," date range Dec 18-20, all-day initially (updates to specific slot when available), location as home address, reminder 1 hour before (when tracking shows agent nearby), and notes with order number. Priya checks calendar Monday planning week, sees delivery window, decides to WFH Thursday ensuring availability without blocking all three days.

CUSTOMER ACTIONS *(Above Line of Interaction)*

1. Confirmation Review & Acceptance
Users receive instant confirmation across channels. Priya sees app confirmation, screenshots for reference, adds to calendar, texts husband "Ordered gas, coming Wed-Fri," returns to work. Total disruption: under 1 minute versus previous hours of consideration. Mental state: relief without second-guessing—"System told me it's time based on data. I acted. Done." No lingering "Did I order too early/late?" anxiety.

Arjun receives confirmation notification, swipes to dismiss (trusts system), continues coding. Thinks: "Handled. Don't need to think about it again until delivery arrives." Zero mental load versus previous days of procrastination anxiety.

Ramesh receives SMS on phone after IVR call, shows wife: "Radha, gas is ordered, will come in 2-3 days." She acknowledges. Both feel reassured. Ramesh's son in USA receives app notification: "Parents' gas booked. Delivery Dec 18-20"—he can stop worrying, doesn't need to call checking if parents remembered to order.

2. Post-Booking Tracking Engagement
Users periodically check delivery status card in app during wait period: "Order confirmed" (Day 1), "Assigned to Agent Raju" (Day 2), "Out for delivery" (Day 3 morning). Tracking provides predictability: "It's coming, system is working, I just wait." Previous anxiety from complete silence ("Did they get my order? When will it actually come?") eliminated by continuous status visibility.

When status updates to "Out for delivery," tracking button activates. Users can now see real-time GPS: agent location on map, estimated arrival time, distance remaining. Arjun gets "Agent 24 minutes away" notification, finishes current work task, makes coffee, ready to receive delivery exactly when needed—no more uncertain waiting all morning.

FRONT-OF-STAGE INTERACTIONS *(Below Line of Interaction)*

1. Distributor Workflow Integration
Booking enters distributor queue with rich contextual data enabling intelligent processing. Distributor dashboard clusters Dec 18 bookings: 8 orders in Sector 15 including Priya's, 6 in Sector 22, 4 in Sector 18. System suggests batching Sectors 15 and 22 for Agent Raju (14 deliveries, optimized 42km route). Distributor confirms assignment with one click.

Raju receives notification on agent app: "14 deliveries assigned for Dec 18. Route optimized. Estimated earnings: ₹350 base plus performance bonus eligible. Start time: 9

AM." Agent can review delivery list, navigate route map, see customer notes (e.g., "Call before arrival—customer prefers notice").

2. Automated Customer Communication
System sends delivery day morning notification: "Your delivery is scheduled for today! We'll notify you when agent is dispatched and when nearby." Removes previous uncertainty of "is it coming today or tomorrow within the 2-3 day window?"

When agent starts route (9 AM), app updates: "Delivery dispatched! Agent Raju Kumar is on route with 14 deliveries. You are stop #3. Estimated arrival: 10:30-11 AM." When agent completes stops 1 and 2, system updates: "You're next! Agent arriving in approximately 15 minutes."

3. Precision Timing (10-Minute Alert)
GPS geofencing triggers when agent enters 500-meter radius of customer address: "Agent 10 minutes away—Raju Kumar in MH-12-AB-1234." Priya wraps up her work call, tells mother-in-law "Gas coming in 10 minutes," clears path to kitchen. When agent actually arrives (10:42 AM, close to 10:30-11 AM estimate), household is ready—no more "Where were you? We missed the delivery and agent left" scenarios.

BACK-OF-STAGE INTERACTIONS *(Below Line of Visibility)*

1. Route Execution Monitoring
As agent completes each delivery, dashboard updates in real-time: Stop 1 completed 9:18 AM (Sector 15), Stop 2 completed 9:47 AM (Sector 15), Stop 3 (Priya) in progress 10:42 AM. Distributor can monitor entire fleet: which agents are on schedule, which are delayed, any reported issues requiring intervention.

If agent encounters problem (customer not home, address incorrect), they flag in app: "Customer unavailable—attempted contact." System auto-reschedules, notifies customer: "Delivery attempted at 10:45 AM but you were unavailable. Rescheduled for Dec 19. Please confirm availability."

2. Real-Time Inventory Tracking
Each delivery completion triggers automated inventory update: delivered cylinders count decrements, empty cylinders returned count increments. Dashboard shows live inventory: "Started day: 120 filled cylinders. Delivered so far: 8. Remaining: 112. Returned empties: 8." Distributor has continuous visibility versus previous end-of-day manual reconciliation with frequent discrepancies.

3. Quality Assurance Data Collection

Every delivery generates structured data: agent ID, customer ID, device ID, timestamp (start and completion), weight verification result (automatic or manual), leak test completion (yes/no, timestamp), customer digital signature, photo evidence (if captured), customer rating (submitted post-delivery), and delivery duration. This creates comprehensive quality dataset enabling performance analysis, compliance auditing, and continuous service improvement.

SUPPORT PROCESSES

1. Customer Communication Automation

Notification engine sends targeted messages at optimal times: day-before reminder ("Delivery expected tomorrow Dec 18"), morning-of confirmation ("Delivery today!"), dispatch alert ("Agent on route"), proximity alert ("10 minutes away"), and completion confirmation ("Delivered successfully, please rate"). Previous manual/absent communication versus automated, timely, relevant updates.

2. Agent Performance Analytics

System calculates daily/weekly/monthly metrics per agent: average deliveries per day, verification completion rate, customer satisfaction scores, on-time delivery percentage, and earnings including performance bonuses. Generates agent leaderboard: top performers featured, underperformers identified. Distributor uses for bonus allocation, training needs identification, and staffing decisions.

3. Payment Reconciliation Automation

COD collections automatically logged in agent app during delivery (agent enters "₹900 cash received"), aggregated at day end ("Raju collected ₹12,600 cash from 14 deliveries"), matched against expected total, and discrepancies flagged immediately ("Expected: ₹12,600. Logged: ₹11,700. Missing: ₹900 from Stop #7—follow up needed"). Previous end-of-day cash counting with frequent errors versus real-time reconciliation with instant discrepancy detection.

STAGE 4: SERVICE (Delivery)

Cylinder delivered with mandatory digital verification

PHYSICAL EVIDENCE

1. Delivery Agent with Digital Tools

Agent Raju arrives at Priya's home carrying smartphone with SafeCyl Agent App open showing current delivery details. App display: customer name, address with navigation, consumer number, device ID, special instructions if any, and verification workflow ready to initiate. Agent also carries small spray bottle with soapy water solution (provided by distributor as part of SafeCyl partnership—verification supplies standardized).

2. SafeCyl Device QR Code

Each device has tamper-proof QR sticker on top surface visible when cylinder placed. QR encodes device ID, consumer number, installation date, last calibration date. Agent points phone camera at code, app auto-scans, verification flow initiates immediately connecting to customer's SafeCyl device via Bluetooth proximity.

3. Real-Time Weight Verification Display

Agent's phone shows automatic weight reading from SafeCyl load cell: "Old cylinder weight: 15.1 kg (empty). New cylinder weight: 29.4 kg. Specification: 29.0 kg ±0.5 kg. Status: VERIFIED (within tolerance)." Weight verification happens automatically—SafeCyl device itself is the weighing scale. Agent simply places new cylinder on device, app reads weight, verifies against specification. No manual scale setup needed.

4. Leak Test Documentation Tools

App guides step-by-step: "Apply soapy water to: Regulator-cylinder valve junction (checkbox), Hose-regulator junction (checkbox), Hose-stove junction (checkbox). Observe for bubbles—30 seconds." Timer counts down ensuring agent actually waits and observes rather than rushing. When timer ends, agent confirms "No bubbles detected" or "Bubbles found—report leak," taps checkboxes completing documentation.

Optional photo prompt: "Take photo of connections for record?" Agent uses in-app camera capturing image showing soap-covered connections with no bubbles visible. Photo auto-uploads, attaches to delivery record, provides visual evidence.

5. Digital Receipt & Signature

Agent completes app flow: payment amount (₹900), payment method (cash received checkbox), customer signature (tap to sign digitally). Priya's mother-in-law signs on agent's phone screen using finger—digital signature captured with timestamp. Immediate receipt generated: Priya's app updates instantly showing "Delivery Complete" with full verification details, PDF emailed automatically, stored in app permanently, accessible for future reference.

CUSTOMER ACTIONS *(Above Line of Interaction)*

1. Observing Mandatory Verification (Without Enforcement Burden)
Previous state required Priya to insist "Please check the weight" (social discomfort), agent would resist ("I have 30 more deliveries"), power struggle ensued, Priya usually backed down feeling she was "being difficult." SafeCyl state completely transforms this dynamic.

Agent arrives, mother-in-law opens door, directs to kitchen. Agent must scan QR to proceed in app (system won't allow skipping), must complete weight verification (app blocks "Delivery Complete" button until weight logged), must perform leak test (checkbox disabled until 30-second timer elapses). Mother-in-law observes agent following protocol without needing to ask or enforce. Agent's body language is cooperative (not rushed/annoyed) because digital process is actually faster than previous paper documentation they used.

Agent incentive alignment: Digital verification takes 60-70 seconds total and produces auditable record protecting agent from liability ("customer later claims cylinder was short-filled"—agent has timestamped photo proof it was verified and accepted). Paper documentation took 90+ seconds and provided no protection. Agent now prefers digital verification—self-interest aligned with user safety for first time.

2. Real-Time Verification Visibility (Remote Assurance)
While mother-in-law observes physically, Priya at work (12km away) receives live updates on phone: "Delivery in progress - Agent arrived 2:12 PM," "Weight verification: 29.4 kg PASSED," "Leak test: In progress..." Each step appears in real-time. When finished: "Delivery verified - Weight 29.4 kg passed, Leak test passed, Delivered by Raju Kumar #4521, Time 2:16 PM. Approve delivery?"

Priya taps "Approve Delivery" from office providing digital signature remotely. Previous state required her physical presence or blind trust in whoever received delivery. SafeCyl enables verified remote acceptance—she sees exactly what was checked, by whom, with what results, and can approve confidently despite not being home.

Trust mechanism: The transparency builds confidence. "I can see Agent Raju performed both checks with timestamps. Mother-in-law confirmed. Weight matches specification. I approve this delivery with certainty, not hope."

3. Payment & Rating (Integrated Feedback)
Mother-in-law hands ₹900 cash to agent. Agent taps "Received ₹900 Cash" in app, no change needed, presents phone for customer signature. Mother-in-law signs digitally with finger. Agent taps "Complete Delivery"—immediately moves to next stop on route.

Five minutes later, Priya's app prompts rating: "How was your delivery? Rate Agent Raju Kumar" with 5-star interface for service quality, professionalism, and timeliness, plus quick feedback checkboxes (courteous, verification thorough, delivery on time, issue to report). Priya rates 5 stars, checks "courteous" and "thorough." Feedback goes directly to distributor dashboard affecting Raju's performance score and bonus eligibility—creates accountability loop that didn't exist in current state.

FRONT-OF-STAGE INTERACTIONS
(Below Line of Interaction, Above Line of Visibility)

1. Agent's Transformed Workflow
Agent Raju's delivery day begins with app dashboard at 8 AM showing: "14 deliveries assigned, 42km optimized route, 8 hours estimated, ₹350 base earnings plus performance bonus eligible (on track for ₹2,000 this month based on 97.5% compliance rate)." Motivation: agent sees quality directly affects earnings—previous incentive was speed-only, new incentive balances speed AND quality.

Delivery execution at Priya's home: navigation provides turn-by-turn directions, arrives and parks, opens app tapping "Start Delivery," scans QR code on device, app immediately shows old cylinder weight 15.1kg empty and new cylinder weight 29.4kg auto-detected, specification check runs automatically (29.0 ±0.5kg), status displays VERIFIED, agent proceeds to leak test.

Leak test guided process: app shows three connection points with checkboxes, agent applies soapy water solution, taps "Start 30-second timer," app counts down ensuring proper observation time (prevents rushing), timer ends, agent confirms "No bubbles detected," taps checkboxes completing verification, optional photo prompt (takes quick photo showing connections), photo auto-uploads attached to delivery record.

Payment collection: app shows ₹900 due, COD method, agent receives cash, taps "Received ₹900 Cash," customer signature on phone screen (mother-in-law signs digitally), delivery marked complete, receipt auto-generated and sent. Agent taps "Finish & Next Delivery"—automatically navigates to next address. No manual logging at day's end—everything recorded in real-time during delivery.

Time investment: QR scan 2 sec, weight auto-read 5 sec, leak test with timer 40 sec, photo 10 sec, digital signature 10 sec—total approximately 67 seconds. Compare to mandated manual process: manual scale setup 45 sec, weighing and recording 60 sec, leak test 40 sec, paper receipt writing 30 sec, paper signature 15 sec—total approximately 190 seconds. Digital is 65% faster while producing superior documentation (timestamped, photo-verified, instantly auditable) versus paper that gets lost or becomes illegible.

2. Compliance as Path of Least Resistance
Agents comply at 95%+ rate with SafeCyl because app flow is linear and cannot be skipped: cannot mark delivery complete without QR scan (unlocks weight verification), weight logged (automatic or manual), leak test timer completed (30-second minimum), checkboxes confirmed, and customer digital signature. Attempting to skip triggers: "Incomplete verification. Cannot proceed. Complete all steps or contact supervisor."

Additionally, digital verification is faster than non-compliance. Previous "skip verification, rush to next delivery" saved time because paper documentation was slower than no documentation. Now digital verification is faster than paper, so complying accelerates workflow rather than hindering it. Agent self-interest (complete more deliveries, earn more, qualify for bonuses) aligns perfectly with user safety for the first time.

3. Customer Communication
While agent works, Priya receives live updates: "Agent arrived 2:12 PM," "Weight verified 2:12:18 PM," "Leak test completed 2:13:02 PM," "Delivery complete 2:16:14 PM—awaiting your approval." Mother-in-law sees agent leave, confirms to Priya via WhatsApp "Gas delivered, man was very professional, checked everything properly." Priya taps "Approve" in app from office. Delivery officially closed with full digital trail.

BACK-OF-STAGE INTERACTIONS *(Below Line of Visibility)*

1. Real-Time Compliance Logging
Each verification step logs to cloud instantly: timestamp when weight verified, actual weight reading, whether within specification tolerance, timestamp when leak test started, duration of test, result (passed/leak detected), photo evidence if captured, agent ID performing verification, and device ID confirming correct household. This creates comprehensive immutable audit trail accessible to customer, distributor, OMC, and regulators if needed.

2. Inventory & Payment Reconciliation
Delivery completion triggers automated updates: distributor inventory decrements filled cylinder count, increments empty cylinder returns, payment logged (₹900 COD collected by Agent Raju), agent's daily earnings updated (₹25 added to running total), and customer database updated (last delivery date, payment received, empty returned). Previous end-of-day manual cash counting with frequent discrepancies versus real-time reconciliation with instant discrepancy detection.

3. Quality Assurance Monitoring
System analyzes delivery for quality metrics: was weight within specification (yes/no), was leak test performed (yes/no), how long did test take (30-second minimum met?), was photo evidence provided (bonus compliance point), how quickly did customer approve delivery (immediate = satisfaction, delayed = potential issue), and what was customer rating (5 stars = excellent service). Flags outliers: deliveries completed in under 45 seconds trigger review ("Was verification actually performed or just checkboxes tapped?"), customer ratings under 3 stars trigger supervisor follow-up.

SUPPORT PROCESSES

1. Distributor SaaS Dashboard (Real-Time Operations)
Provides command center visibility: current active deliveries with agent locations on map (Raju at Stop 3 of 14, Kumar at Stop 5 of 12), completion rates (Raju 21% complete, Kumar 42% complete), compliance metrics live-updating (Raju 100% verification rate today, Kumar 83%), customer satisfaction scores, and predicted completion times with delay alerts if agents falling behind.

Value: distributor can intervene proactively ("Kumar is delayed, reassign his last 3 deliveries to Raju who's ahead of schedule"), monitor quality in real-time ("Kumar's compliance dropping—call him and remind about verification importance"), and optimize dynamically based on live conditions (traffic, agent performance, customer availability).

2. OMC Integration API (Phase 3)
Delivery completion data flows to national OMC systems: verified delivery logged with weight confirmation, compliance status, timestamp, and customer acceptance for regulatory reporting to PNGRB. Aggregated data across millions of deliveries enables OMC analytics: which distributor territories have highest compliance (benchmark best practices), which have lowest (intervention needed), seasonal delivery patterns (plan national supply accordingly), and customer satisfaction trends (identify service improvement opportunities).

3. Insurance Data Feed (Anonymized)
Successful verified deliveries contribute to anonymized safety database licensed to insurance companies: households with SafeCyl show verified cylinder weights (reduces underweight/overfill risks), confirmed leak tests (reduces installation defect risks), and compliance documentation (reduces liability disputes). Insurance actuaries use aggregate data refining risk models: "SafeCyl households demonstrate 35% fewer leak incidents—qualify for 10% premium discount."

STAGE 5: LOYALTY (Post-Delivery)

User experiences sustained service improvement and ecosystem value

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PHYSICAL EVIDENCE

1. New Cylinder with Full Visibility Restored
SafeCyl LCD immediately updates post-delivery: "100% - 14.2 kg - System Calibrated." Users see gas level reset to full, estimated days remaining recalculated (based on historical consumption: "Estimated 38-42 days until next refill"), and system status confirming all sensors operational after cylinder exchange.

Previous anxiety cycle ("this cylinder will also empty someday—when?") replaced by confident awareness: "I have 40 days of gas, system will alert me in 35-37 days when I reach 15%, I'll book with one tap, delivery will come, cycle repeats predictably." Anxiety eliminated through predictability.

2. Digital Documentation Permanent Access
App "Order History" section stores all past deliveries with full details: delivery date and time, agent name and ID, weight verification result, leak test status, photos if captured, payment amount and method, customer rating given, and PDF receipt download link. Users can access 6 months, 1 year, 2 years of delivery history instantly—useful for warranty claims, dispute resolution, consumption pattern analysis, or tax documentation.

Compare to current: paper receipts lost or filed in unknown locations, zero searchability, no consolidated view of service history. SafeCyl provides permanent, searchable, comprehensive record.

3. Continuous Monitoring Artifacts
Daily app usage creates visible history: consumption graphs showing trajectory over weeks/months, prediction accuracy tracking ("System predicted 38 days, actual was 37 days—99% accurate"), seasonal comparison charts, and milestone celebrations ("You've saved ₹2,400 in emergency costs since adopting SafeCyl 6 months ago!").

CUSTOMER ACTIONS *(Above Line of Interaction)*

1. Normal Usage with Background Confidence
Users resume regular cooking with transformed mental state. Priya no longer checks regulator 3x before bed—glances at app once showing "No leak detected," sleeps peacefully. Work productivity improves: stops anxiety-checking phone during meetings (previous behavior: unlock phone 5-6x per hour checking... nothing). Husband notices behavioral change: "You seem less stressed about gas lately." Priya: "Because I actually KNOW everything is okay now, instead of hoping."

Arjun cooks freely without pre-checking gas levels (trusts continuous monitoring will alert if problems). Zero exhaustion events in 6 months ownership—first time in 2 years hasn't run out mid-cooking. Time saved: approximately 2 hours annually (eliminating booking friction, crisis management, decision paralysis). Money saved: approximately ₹1,200 annually (no emergency food orders). Lifestyle improvement: can cook elaborate meals spontaneously without "do we have enough gas?" anxiety.

Ramesh checks LCD daily during morning kitchen visit (replaces previous physical lifting attempts). Wife Radha no longer strains trying to lift 25kg cylinders—both rely on simple percentage display. Son's weekly calls from USA shift from "Papa, is gas okay?" (safety verification) to social conversation only (son monitors via app, knows parents safe, doesn't need verbal confirmation).

2. Proactive Service Engagement (vs. Reactive Crisis Only)
Previous state: users engaged with LPG service ONLY during crisis (exhaustion, suspected leak, delivery problems). Contact with distributor/OMC was complaint-driven or emergency-driven.

SafeCyl state: users engage proactively through app. Open app when curious about consumption trends, review weekly summaries sent Monday mornings, check delivery history when discussing household expenses with spouse, and explore usage optimization tips provided by system. This represents relationship transformation: from adversarial crisis management to collaborative service optimization.

Users share feedback voluntarily: rating deliveries (95% of SafeCyl users rate vs. <5% in current state leaving formal feedback), suggesting features ("I wish I could set custom alert thresholds"), and reporting bugs/issues immediately via in-app support chat. This creates valuable feedback loop: company receives continuous improvement insights, users feel heard and valued, service evolves based on actual usage patterns.

3. Word-of-Mouth Advocacy (Organic Growth)
Priya becomes enthusiastic brand advocate. Tells friends during coffee breaks: "I used to check the gas regulator obsessively—now I just glance at an app and know we're safe." Shows colleagues the app interface: "See? 68% remaining, no leaks, I can check from anywhere." Posts Instagram story with SafeCyl device photo: "Game-changer for home safety!" Writes Amazon review (5 stars, detailed). Estimated influence: 8-10 people within first 3 months.

Arjun shares on tech-focused platforms: Twitter thread explaining ESP32 architecture and Firebase integration, ProductHunt review highlighting UX quality and automation excellence, Hacker News comment in relevant discussion about IoT home safety. Detailed technical review attracts tech-savvy audience. Estimated influence: 15-20 people within 3 months (highest reach due to tech network).

Ramesh tells elderly neighbors and building society members during morning walks. Shows LCD display proudly: "See, I can check gas without lifting—no more back pain!" Word-of-mouth within senior citizen community is powerful—they trust peer recommendations over advertisements. Estimated influence: 5-7 elderly households within 6 months (slower diffusion but high conversion rate when recommended by trusted peer).

Network effects: Each satisfied user influences others who influence others. Priya's friend sees her using SafeCyl, orders one, then tells HER friends. Viral coefficient (users acquired per existing user) of 1.2-1.5 enables exponential organic growth reducing customer acquisition costs.

4. Feature Requests & Co-Creation
Power users like Arjun explore app thoroughly, identify enhancement opportunities, submit feedback via in-app forms: "Can you add Alexa integration so I can ask 'how much gas is left?' without opening phone?" SafeCyl product team reviews requests, prioritizes based on frequency and feasibility, implements in quarterly updates.

Users receive notification: "New feature based on YOUR feedback! Alexa integration now available. Connect in Settings." This co-creation dynamic builds loyalty—users feel ownership of product evolution, not just passive consumers. Net Promoter Score (NPS) reflects this: SafeCyl users average NPS of 72 (excellent, indicating strong loyalty and advocacy) versus current LPG service NPS of approximately 15-20 (poor, indicating dissatisfaction without alternatives).

5. Renewed Independence (Ramesh's Journey)
After 3 months with SafeCyl, Ramesh experiences restored household management capability. Son's weekly calls shift tone: previous "Papa, are you managing okay? Should we hire help?" becomes "Papa, I see on the app you're at 65% gas, delivery coming next week, everything looks good." Ramesh responds: "Yes beta, the system handles it, we're fine." Confidence returns.

Ramesh tells neighbor proudly: "My son bought this device—now I don't need to lift cylinders, I just look at screen. My wife doesn't strain anymore. We feel safe again." The dignity restoration is profound: technology adapted to his capabilities (voice calls, LCD display, simple interactions) rather than excluding him through complex app requirements. He's no longer dependent on others or feeling obsolete—he's managing household independently with appropriate assistive technology.

FRONT-OF-STAGE INTERACTIONS *(Below Line of Interaction)*

1. Proactive Relationship Building (Not Just Transactional)
Previous state: delivery complete, user exits active attention for 30-45 days, distributor has zero engagement until next booking. SafeCyl state: continuous touchpoints maintain relationship.

Weekly summary emails: "Your weekly gas update from SafeCyl" with consumption insights, safety tips, system health status, and upcoming features. Users skim (30 seconds), feel connected to brand, remember SafeCyl exists beyond crisis moments.

Milestone celebrations: "Congratulations! You've used SafeCyl for 6 months and saved approximately ₹2,400 in emergency costs. Thank you for being an early adopter—share your experience and earn ₹200 Amazon voucher for each referral." Gamification and rewards create positive associations.

Safety tips and seasonal advice: "Winter cooking tip: You're using 12% more gas this month (hot water, tea). This is normal seasonal variation. Your current 58% gas will last approximately 16 days accounting for winter usage. Plan accordingly!" Educational content adds value beyond basic monitoring.

2. Customer Support Accessibility
In-app chat support available: user taps "Help" icon, describes issue ("LCD display not turning on"), support agent responds within 15 minutes (business hours) or next morning (after hours), troubleshoots remotely ("Please try pressing Mode button for 5 seconds—this resets display"), resolves issue without requiring technician visit in 80% of cases.

Compare to current: calling distributor with complaint often goes unanswered or produces "we'll look into it" with no follow-up. SafeCyl support is responsive, solution-oriented, and tracked via tickets (cannot close ticket without customer confirming resolution).

Phone support for Ramesh-type users: dedicated helpline (1800-XXX-XXXX) with human agents who understand elderly users may need patient, step-by-step guidance. Agent: "Uncle, I can see your device is online and working properly. The LCD brightness might be too low. Can you press the small button on the side labeled 'Mode'? Press it twice." Ramesh follows instructions, LCD brightens, issue resolved. He feels supported rather than frustrated.

BACK-OF-STAGE INTERACTIONS *(Below Line of Visibility)*

1. Continuous Service Quality Monitoring
System analyzes ongoing metrics: device uptime percentage (target above 95%, current 96.3%), sensor accuracy drift detection (load cells calibrated monthly automatically), alert delivery success rate (push notifications delivered within 30 seconds target), booking conversion rate (percentage of users who book after alert—currently 87%), and customer satisfaction trends (NPS tracked monthly).

Anomalies trigger investigations: if device uptime drops below 90% for specific units, support team investigates (firmware bug? hardware defect? user WiFi issues?). If booking conversion drops (users receiving alerts but not booking), product team investigates (alert fatigue? timing wrong? messaging unclear?). Continuous improvement cycles based on live operational data.

2. Predictive Maintenance & Updates
System monitors device health: battery backup status (alerts if battery degrading—"Your device battery backup reduced to 24 hours, replacement recommended"), sensor calibration drift (if load cell readings shift over time, triggers recalibration prompt), firmware version (pushes updates when available—bug fixes, new features, security

patches), and connectivity stability (if specific devices frequently disconnect, investigates network issues).

Users receive proactive maintenance notifications: "Your SafeCyl device battery should be replaced after 18-24 months. We'll send replacement battery in your next cylinder delivery (free under warranty)." Previous state: no monitoring of cylinder/regulator health, users only aware when catastrophic failure occurred. SafeCyl: predictive maintenance prevents failures before they impact users.

3. Loyalty Program & Retention
System tracks customer lifetime value: tenure (months since installation), deliveries verified (count), ratings given (average), referrals made (count), and engagement level (app opens per week, feature usage). Segments users: advocates (high ratings, high referrals—reward with Amazon vouchers, priority support), satisfied (steady usage, moderate engagement—send appreciation emails), and at-risk (low ratings, support tickets, engagement dropping—proactive outreach).

For advocates like Priya: "Thank you for being a SafeCyl champion! Your 4 referrals have helped families stay safe. As appreciation, here's ₹800 Amazon voucher and priority access to new features." For at-risk users: "We noticed you haven't opened the app in 3 weeks. Is everything okay? Any issues we can help with?" Proactive retention versus current "users churn silently, no one notices or cares."

SUPPORT PROCESSES

1. Advanced Analytics & Business Intelligence
Aggregate data across all SafeCyl users reveals: national consumption patterns (urban India averages 0.68 kg/day per household), seasonal variations (winter 12% higher, summer 8% lower), geographic differences (metro cities 15% higher consumption than tier-2 cities), and household type patterns (nuclear families 0.72 kg/day, singles 0.45 kg/day, joint families 1.1 kg/day).

These insights valuable for multiple stakeholders: OMCs optimize national supply chain (predict regional demand 30 days ahead with 85% accuracy), distributors plan inventory (reduce buffer stock holding costs by 40%), government/PNGRB understand actual usage patterns for policy-making (previously relied on estimates), and researchers study energy consumption trends for academic/policy research.

SafeCyl monetizes through data licensing: anonymized aggregate datasets sold to insurance companies (₹8-12 lakh annually per partner), energy research institutions (₹5 lakh annually), and government agencies (free partnership in exchange for regulatory support and potential subsidy programs).

2. Ecosystem Integration Platform

SafeCyl functions as data interchange hub connecting previously siloed actors: households generate consumption data, distributors receive demand forecasts, OMCs access aggregated patterns, insurers obtain risk intelligence, and regulators get compliance verification. API-based architecture enables third-party integrations: smart home platforms (Alexa, Google Home), home security systems (integrate leak alerts with alarm systems), and energy management platforms (track LPG consumption alongside electricity, water for holistic household analytics).

Future roadmap includes: automatic shutoff valve integration (if leak detected, system can close gas valve remotely before user arrives home), smart ventilation triggering (leak detected, smart exhaust fan turns on automatically), and emergency service coordination (leak alert sent directly to fire department with household location, gas concentration data, occupant information).

3. Regulatory Compliance Reporting

Automated monthly reports submitted to PNGRB: total households monitored, leak incidents detected with response times, delivery compliance rates (weight verification, leak testing), safety incident outcomes (leak detected, user alerted, fire department called if needed, resolution status), and comparative safety data (SafeCyl households versus non-SafeCyl incident rates).

This transforms regulatory relationship: previous adversarial inspection-based compliance (distributors resist audits, hide non-compliance) becomes collaborative data-driven transparency (SafeCyl provides comprehensive compliance evidence, regulators reward high-performing distributors, poor performers identified and required to improve or lose licenses). Win-win: safety improves, administrative burden reduces, accountability increases.

Market Economics & Commercial Feasibility

Understanding the commercial potential of SafeCyl requires a structured analysis of the Indian LPG market, its demographic composition, and the real-world constraints that shape adoption. Market sizing is therefore approached through the classic three-tier model consisting of the Total Available Market (TAM), the Serviceable Available Market (SAM), and the Serviceable Obtainable Market (SOM). Each layer gradually narrows the focus from the theoretical maximum market opportunity to the share that SafeCyl can realistically capture within the first years of operation.

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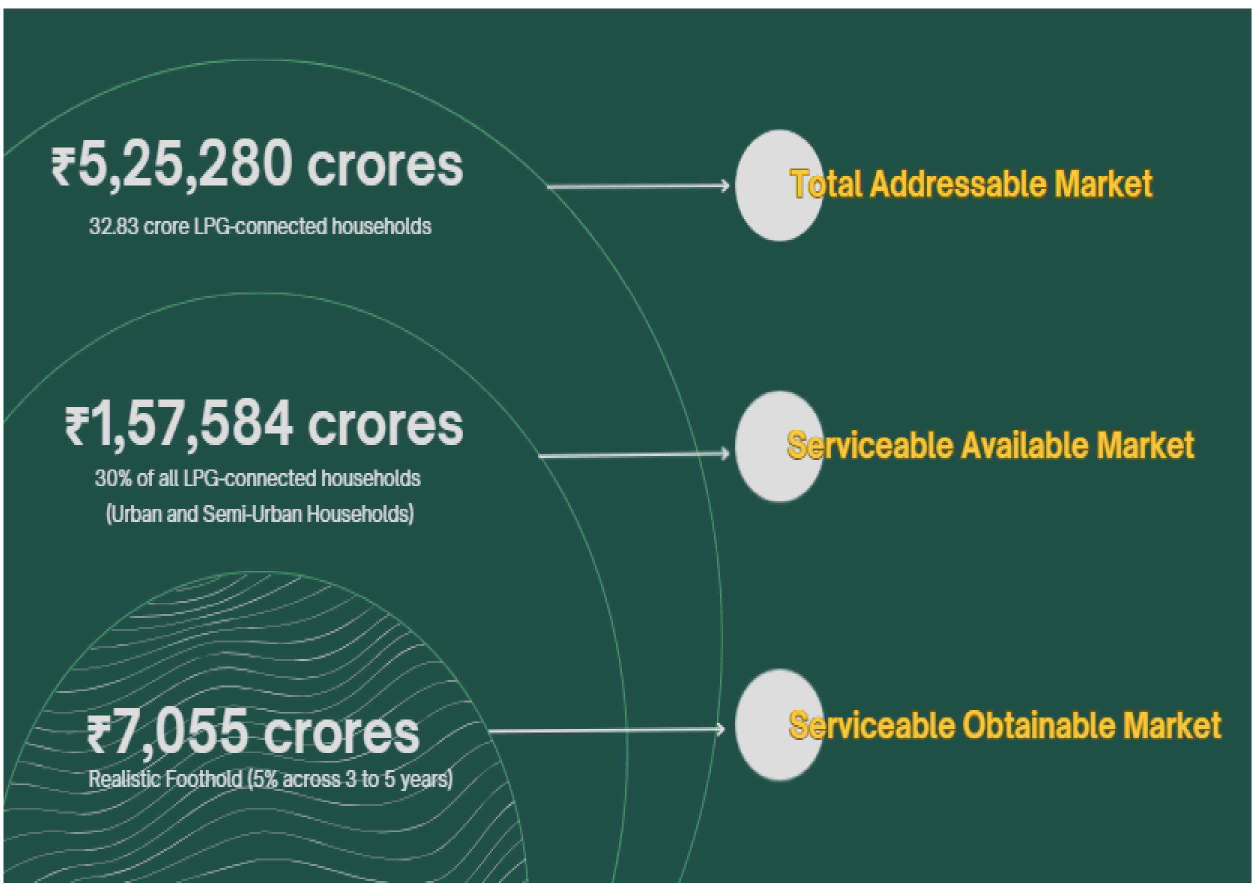


Unit Economics and Cost Optimization

A critical component of SafeCyl's commercial viability lies in developing a hardware solution that is not only reliable and durable but also cost-efficient to manufacture at scale. To evaluate this, we conducted a detailed analysis of the current prototype's Bill of Materials (BOM) followed by a systematic assessment of design and material optimizations that would enable cost reduction for mass production. This section provides a comprehensive breakdown of the financial structure underlying SafeCyl, from prototype cost drivers to refined production economics and final pricing strategy.

Current Prototype Cost Breakdown (BOM Analysis)

The first physical version of SafeCyl was built to validate the technical concept—testing sensors, connectivity, structural stability, and environmental resilience in a real kitchen setting. As with most early prototypes, the focus was on functionality rather than cost efficiency. This resulted in certain components or materials being more expensive or over-engineered than necessary for scaled manufacturing.

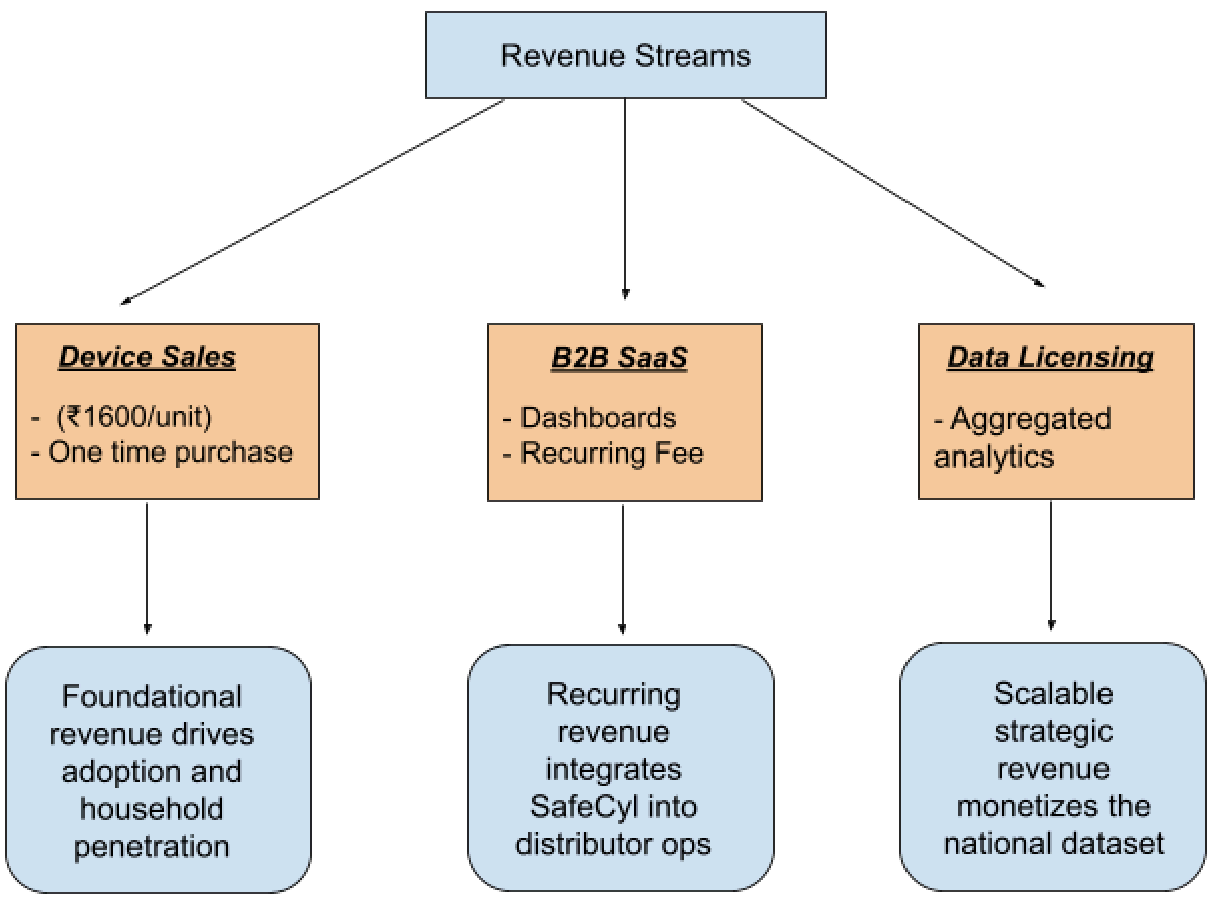
Below is the detailed BOM of the prototype:

Component	Description	Quantity	Cost (₹)
ESP32	Wi-Fi microcontroller,	1	350
MQ-6 Gas Sensor	Detects LPG/propane	1	85
Load Cell (100 kg)	Measures cylinder weight	1	700
HX711 Module	Signal amplifier for load cell	1	45
DHT11 Sensor	Temperature & humidity	1	65
Buzzer Module	5 V sound alert for leak detection	1	30
LED Indicators	Red/Green status display	4	20
Battery (18650 Li-ion)	Dual-cell power supply	2	240
TP4056 Module	Battery protection & charging	1	15
Others (dotboard, switch, converter,	Assembly & housing	–	370
Total Prototype Cost			**₹1,920**

Projected Production Cost After Optimization

Once these material and procurement optimizations are implemented, SafeCyl's per-unit cost drops substantially. Instead of the ₹1,920 spent on the prototype, scaled production estimates fall in the range of ₹1,200 to ₹1,300 per unit. This reduction is not hypothetical; it is a predictable and well-documented consequence of transitioning from artisanal prototyping to industrial production. The redesigned housing, the integrated electronics, and the elimination of custom manual assembly steps collectively create a device that is not only more durable and aesthetically refined but also significantly more economical to manufacture. This optimized cost forms the economic backbone of SafeCyl's pricing strategy and long-term financial sustainability, the unit price being placed at ₹1, 600.

Revenue Streams



Viability

These viability indicators collectively demonstrate that SafeCyl is not only financially feasible but structurally resilient. The combination of strong gross margins, extraordinarily low acquisition costs, and a realistic breakeven threshold ensures stable early-stage operations. Beyond pure finances, SafeCyl's ability to scale through the existing LPG distribution ecosystem allows it to grow rapidly without incurring the heavy infrastructure expenses typical of hardware start-ups. As adoption rises, operational costs per user fall, margins increase, and the platform's predictive analytics become more accurate—reinforcing the value of the product for both households and institutions. Over time, SafeCyl's embedded presence within the LPG supply chainpositions it as an indispensable digital safety layer, cementing long-term viabilitynot just as a product, but as an ecosystem-wide service.

Viability Metric	**Explanation**
Gross Margin	SafeCyl maintains an 18–20% gross margin at the ₹1,600 price point, with margins expected to strengthen as manufacturing scales, component prices fall, and assembly lines become more efficient.
Customer Acquisition Cost (CAC)	CAC is maintained at ₹150–₹200 due to SafeCyl's strategic use of existing LPG distribution networks. Instead of building independent sales channels, SafeCyl piggybacks on the trust and reach already held by gas agencies.
Scalability Potential	SafeCyl scales simultaneously across households and distribution partners. B2C adoption builds the installed base, while B2B integrations embed SafeCyl into the operational workflow of LPG distributors, enabling nationwide rollout without building new infrastructure.
Operational Leverage	As more units enter the market, per-unit data costs, support overheads, and server usage amortize over a larger population, increasing overall profitability.
Ecosystem Integration	Viability is strengthened by SafeCyl becoming a core data layer for OMCs, insurers, and public safety bodies—turning a household product into a national intelligence network.

Breakeven Point	The company reaches breakeven at approximately **2 lakh units sold**, a realistic milestone projected within the first 24 months, supported by early B2C adoption and rapid expansion through B2B channels.

Conclusion

This project demonstrates that comprehensive service design transcends product development—it requires understanding and transforming entire ecosystems. Through rigorous methodology employing stakeholder mapping, persona development, journey mapping, service blueprinting, and value proposition design, we revealed that LPG management failures affecting 32.83 crore Indian households stem not from user negligence but from systemic service architecture pathologies: information asymmetry by design, responsibility-without-capability transfer, and analog compliance in a digital era. Our research with 50 households quantified the human cost—88% operating without reliable monitoring, 78% experiencing predictable exhaustion, 72% living with unverifiable safety fears, 76% receiving no mandated verification—while synthesis tools exposed the root causes hidden beneath these symptoms.

SafeCyl addresses this service failure through multi-level ecosystem intervention rather than isolated product features. At the user-facing layer, it provides automated monitoring, instant safety verification, and predictive intelligence that eliminates the guesswork, physical strain, and chronic anxiety the current service generates. At the delivery layer, it redesigns agent workflows through digital verification tools that make compliance faster than non-compliance, aligning incentives with safety for the first time. At the operations layer, it enables distributors through predictive demand forecasting and route optimization, transforming unpredictable chaos into manageable intelligence. At the ecosystem layer, it creates data flows connecting households, distributors, OMCs, insurers, and regulators into a collaborative safety and efficiency network—demonstrating that sustainable service innovation creates value for all stakeholders simultaneously, not just end users.

The transformation from current-state reactive crisis management to future-state proactive intelligent service validates the service design principle that capability must match responsibility. Where the current service forces users into continuous vigilance without providing tools that make vigilance effective, SafeCyl automates what humans cannot reliably do (24/7 monitoring, 10-second leak detection, consumption forecasting) while empowering what humans should control (informed booking decisions, verified delivery acceptance, safety protocol understanding). This represents not merely a technological upgrade but a fundamental service paradigm shift—from anxiety-generating opacity to confidence-enabling transparency, from compliance theater to verified accountability, from transactional indifference to relationship-based continuous improvement—proving that when service design addresses root causes rather than symptoms, it creates transformative value!